

RESIDUAL PROPERTIES OF FINITELY GENERATED GROUPS IN THE WEIHRAUCH LATTICE

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ABSTRACT. Consider, on the space of marked groups, the map $\text{Res}_{\mathcal{C}}$ which associates to a marked group its greatest residually- $\mathcal{C}$ quotient, for different sets $\mathcal{C}$ of groups. Except for trivial cases, this map is discontinuous. We use the Weihrauch lattice to quantify how discontinuous it is. We show that equational noetherianity of $\mathcal{C}$ and whether the set of residually- $\mathcal{C}$ groups is a quasivariety both can be characterized in terms of the position of $\text{Res}_{\mathcal{C}}$ within the Weihrauch lattice. We give exact classifications of $\text{Res}_{\mathcal{C}}$, for $\mathcal{C}$ one of: the set of finite groups, of nilpotent groups, of k -nilpotent groups, $k \geq 1$, of finitely presentable groups, of LEF groups, of torsion free groups.

1. INTRODUCTION

Let $\mathcal{C}$ be a class of groups.

A group G is residually- $\mathcal{C}$ if for every non-identity element g of G there is a quotient of G in $\mathcal{C}$ in which g has a non-identity image.

Every group G has a greatest residually- $\mathcal{C}$ quotient, obtained by taking the quotient of G by the intersection of all kernels of onto homomorphisms from G to groups of $\mathcal{C}$, i.e., by the following normal subgroup:

$$\bigcap_{f: G \twoheadrightarrow H, H \in \mathcal{C}} \ker(f).$$

We denote by $\text{Res}_{\mathcal{C}}(G)$ the group defined this way, and by $\text{Res}_{\mathcal{C}}$ the map sending G to this quotient.

Note that, even though proving the existence of the group $\text{Res}_{\mathcal{C}}(G)$ is trivial, the definition given above provides us with very little information. Indeed, “the set of all kernels of onto homomorphisms from G to groups in $\mathcal{C}$ ” always exists, but it may be very hard to characterize or understand this set.

In some sense, the existence of the group $\text{Res}_{\mathcal{C}}(G)$ is non-constructive. (This statement could be made precise in various ways. For instance [Bau00] provides a constructive framework which coincides with the Weihrauch analysis of problems.)

In the present paper, we provide a topological analysis of the map $\text{Res}_{\mathcal{C}}$ that gives a way to quantify *how non-constructive* is the existence of $\text{Res}_{\mathcal{C}}(G)$.

One of our motivations comes from the study of computability. We will not insist on this aspect, but we quote here a result of Slobodskoi.

The present work stems from a question raised by Vincent Guirardel during the defense of my PhD thesis. It took me several years to understand the question and its relevance. Thanks are also due to Vasco Brattka, whose explanations have greatly improved my understanding of Weihrauch reducibility. Most of this work was conducted while the author was supported by an Alexander von Humboldt fellowship.

Theorem 1.1 (Slobodskoi, [Slo81]). *There exists a finitely presented group whose greatest residually finite quotient is not recursively presented.*

This result, and other related results from [Rau22, BPR25], all revolve around leveraging the discontinuity of the map $\text{Res}_\mathcal{C}$ to obtain computability theoretical results. Our hope is that the present study in turn provides a dichotomy between classes $\mathcal{C}$ which give rise to “a sufficiently discontinuous $\text{Res}_\mathcal{C}$ ”, so that certain computability theoretical phenomena occur, and the other ones, for which the map $\text{Res}_\mathcal{C}$ is too tame to permit these computability theoretical phenomena.

In order to have a nice topological framework to work with, we will study the map $\text{Res}_\mathcal{C}$ on the set of marked groups. A k -marked group is a pair (G, S) , where G is a finitely generated group and $S \in G^k$ is a generating tuple of G . A morphism of k -marked groups $(G, (s_1, \dots, s_k)) \rightarrow (H, (s'_1, \dots, s'_k))$ is a group morphism $f: G \rightarrow H$ which satisfies $f(s_i) = s'_i$ for all $1 \leq i \leq k$.

The marked quotient relation

$$(G, S) \succeq (H, S') \iff \exists f: (G, S) \rightarrow (H, S')$$

is easily seen to be an order relation. The set of k -marked groups forms a poset which is in fact a complete lattice isomorphic to the lattice of normal subgroups of a rank- k free group. In this context, we have the equality

$$\text{Res}_\mathcal{C}(G, S) = \sup \{ (H, S') \in \mathcal{C} \mid (G, S) \succeq (H, S') \}.$$

The set of marked groups is naturally equipped with a Polish topology, the resulting topological space is called the *space of marked groups*.

The main problem of this paper is the following one:

Problem A. What properties of $\mathcal{C}$ can be read off of the topological properties of $\text{Res}_\mathcal{C}$?

We could ask for instance: when is $\text{Res}_\mathcal{C}$ continuous, when are preimages of open sets Σ_2^0 -sets, and so on.

In order to obtain finer classifications than these, we use the notion of continuous Weihrauch reduction, which can be thought of as a generalization of Wadge reduction adapted to functions. We discuss Weihrauch reduction in more details in Section 1.2 of this introduction.

1.1. The logical point of view on the map $\text{Res}_\mathcal{C}$. Another point of view on the map $\text{Res}_\mathcal{C}$ follows the lines of [Cha].

A relation in a marked group (G, S) is an element of the group $\mathbb{F}_S$, the free group over S , which defines the identity in G . A subset of $\mathbb{F}_S$ is the set of relations of a marked group if and only if it is a normal subgroup of $\mathbb{F}_S$.

Thus we can say that a set $R \subseteq \mathbb{F}_S$ of relations *implies* another relation $w \in \mathbb{F}_S$ when w belongs to the normal subgroup generated by R .

When focusing on groups in a certain class $\mathcal{C}$, it is natural to consider implication modulo $\mathcal{C}$:

Definition 1.2. Let R be a subset of a free group $\mathbb{F}_S$ and w an element of $\mathbb{F}_S$. We say that the relations of R *imply w relative to $\mathcal{C}$* if every marked group in $\mathcal{C}$ which satisfies the relations of R also satisfies $w = 1$.

The marked group $\text{Res}_\mathcal{C}(G, S)$ is the marked sgroup whose relations are exactly those that are consequences relative to $\mathcal{C}$ of the relations of (G, S) .

Thus the study of the map $\text{Res}_{\mathcal{C}}$ can be understood as the problem of understanding how hard it is, starting with a set of relations which is sound with respect to the usual implication between relations, to recover a set of relations which is sound with respect to the implication rule induced by $\mathcal{C}$.

1.2. Weihrauch reduction. An informal definition of continuous Weihrauch reducibility is the following:

$$f \leq_W g$$

if and only if f can be computed by “doing something continuous, applying g , and then doing something continuous again”. What makes this informal statement non-trivial to formalize is the fact that what we mean by “something continuous” is in fact a *continuous multi-function*, instead of being simply a function, and that the notion of continuous multi-function used to define Weihrauch reduction is set in the category of *represented spaces* instead of the category of topological spaces.

We will explain this in more details below.

Let us first give some examples to motivate the introduction of Weihrauch reduction.

For X and Y Polish spaces, let us call a function $f: X \rightarrow Y$ Σ_n^0 -measurable if the preimage of an open set by f is Σ_n^0 .

Also, let Lim denote the limit map on Baire space:

$$\text{Lim} : \subseteq \mathbb{N}^{\mathbb{N}} \rightarrow \mathbb{N}^{\mathbb{N}}, \langle p_0, p_1, p_2, \dots \rangle \xrightarrow{n \rightarrow \infty} \lim p_n,$$

where $\langle \rangle$ denotes a bijection between $\mathbb{N}^{\mathbb{N}}$ and $(\mathbb{N}^{\mathbb{N}})^{\mathbb{N}}$. The domain of Lim is the set sequences of $\mathbb{N}^{\mathbb{N}}$ which encode via $\langle \rangle$ a converging sequence. Also, we denote by $\text{Lim}^{(n)}$ the n -fold composition of Lim .

Theorem 1.3 ([Bra04]). *A function $f: X \rightarrow Y$ between Polish spaces is Σ_n^0 -measurable if and only if*

$$f \leq_W \text{Lim}^{(n)}.$$

In other words, $\text{Lim}^{(n)}$ is Σ_n^0 -measurable-complete. Quoting from [BGP21], “Weihrauch reducibility offers a refinement of the Borel hierarchy very much in the same way as many-one reducibility yields a refinement of the Kleene hierarchy”.

The Weihrauch lattice has a rich structure, with algebraic operations on problems, closure operators that capture other reducibility notions [BGP21]. Many types of behaviors of discontinuous functions can be described thanks to a classification in the Weihrauch lattice.

Let us come back to a formal definition of Weihrauch reduction.

A *represented space* is a set X equipped with a partial surjection from Baire space $\delta : \subseteq \mathbb{N}^{\mathbb{N}} \rightarrow X$. When $\delta(p) = x$, we say that p is a δ -name of x . Let $f: X \rightarrow Y$ be a function between represented spaces (X, δ) and (Y, τ) . A *realizer* of f is a partial function $F : \subseteq \mathbb{N}^{\mathbb{N}} \rightarrow \mathbb{N}^{\mathbb{N}}$ such that

$$\forall p \in \text{dom}(\delta), f(\delta(p)) = \tau(F(p)).$$

In words: F maps the name of a point in X to a name of its image.

A multi-function $f : \subseteq X \rightrightarrows Y$ is simply a relation $R \subseteq X \times Y$ which we treat as a multi-valued function via the formula $f(x) = \{y \in Y \mid R(x, y)\}$. A *realizer* of a multi-function f is a partial function $F : \subseteq \mathbb{N}^{\mathbb{N}} \rightarrow \mathbb{N}^{\mathbb{N}}$ such that

$$\forall p \in \text{dom}(\delta) \cap \delta^{-1}(\text{dom}(f)), \tau(F(p)) \in f(\delta(p)).$$

In words: F maps the name of a point in X to a name of one of its images.

A (multi-)function is called *continuously realizable* if it has a continuous realizer.

Definition 1.4 (Continuous Weihrauch reducibility). Let $f : \subseteq X \rightrightarrows Y$ and $g : \subseteq Z \rightrightarrows W$ be multi-functions between represented spaces. We say that f *continuously Weihrauch reduces to* g , and write

$$f \leq_W g,$$

if there is a represented space V and two continuously realizable multi-functions $h : \subseteq V \times W \rightrightarrows Y$ and $k : \subseteq X \rightrightarrows V \times Z$ such that

$$\forall x \in X, \forall (v, z) \in k(x), f(x) \in h(v, g(z)).$$

The role of the space V and of the element v in the above is to “pass the input of f through g ”.

If $f \leq_W g$ and $g \leq_W f$, we say that f and g are *Weihrauch equivalent*, and write $f \equiv_W g$.

1.3. Admissible representations of topological spaces. The definition of Weihrauch reduction given above is set in the category of represented spaces. We can express Weihrauch reduction for functions between topological spaces by establishing a correspondence between some topological spaces (here it will be sufficient to deal with T_0 second countable spaces) and some representations (the *admissible representations*).

First, to any representation ρ of a set X we can associate a topology, the *final topology of the representation*, given by

$$U \subseteq X \text{ is open} \iff \rho^{-1}(U) \text{ is open in Baire space.}$$

Secondly, we associate representations to topological spaces.

Let ρ and τ be two representations of a set X . We say that ρ (*continuously*) *translates to* τ if the function

$$\text{id}_X : (X, \rho) \rightarrow (X, \tau)$$

has a continuous realizer. Two representations of a set X are called equivalent if each one translates to the other one.

Definition 1.5 ([KW85, Sch02]). Let X be a topological space. A representation $\rho : \subseteq \mathbb{N}^{\mathbb{N}} \rightarrow X$ of X is *admissible* if and only if

- It is continuous,
- Any continuous representation of X translates to ρ .

Any second countable T_0 space can be equipped with a *standard representation* (Definition 3.5), which is always admissible, and admissible representations are unique up to equivalence. And we have:

Theorem 1.6 ([KW85]). *Let X and Y be second countable T_0 spaces. Then a function $f : X \rightarrow Y$ is continuous if and only if it is continuously realizable with respect to any admissible representations of X and Y .*

In other words, we have an equivalence of categories between second countable T_0 spaces with continuous functions as morphisms on the one hand, and represented spaces equipped with equivalence classes of standard representations on the other hand.

Moving to the category of represented spaces has two main advantages: computability theory can easily be developed (and this was the original motivation of Kreitz and Weihrauch [KW85]), and continuous multi-functions can be very naturally defined, in a way that seems hard to do in the category of topological spaces (see [BH94] for more details on this).

By results of Schröder [Sch02, Sch21], admissible representations can be constructed exactly for those T_0 topological spaces whose sequentialization is the quotient of a second countable space. In the present article, we will not need this generality.

1.4. Representations of marked groups. The back and forth between represented spaces and topological spaces can be thought of in the following way: each type of description of the elements of a set X yields a topology on X (the sets of open sets being the set of “recognizable properties”), and conversely each (second countable T_0) topology yields a way to describe the elements of X (an element is described by the basic open sets to which it belongs).

In this setting, the topology of the space of marked groups arises in a very natural way: it is the topology associated to marked groups defined by their word problem. In Section 5, we define a representation of the set of marked group by saying that “the name of a marked group is a sequence which encodes the characteristic function of its set of relations”, and the final topology of this representation is precisely the topology of the space of marked groups.

Another topology that is used several times in the present paper is the Scott topology on the set of marked groups, which admits as a base the set of basic sets of the form “those marked groups which satisfy the relations $\{r_1, \dots, r_n\}$ ”, for any finite set of relations $\{r_1, \dots, r_n\}$. This topology is the final topology of the representation associated to *group presentations*: a marked group is described by a (countably infinite) presentation.

1.5. Results. We denote by $\mathcal{RC}$ the set of residually- $\mathcal{C}$ marked groups.

Proposition B. As soon as $\mathcal{RC}$ is neither all marked groups nor the singleton containing the trivial group, the map $\text{Res}_{\mathcal{C}}$ is discontinuous.

Theorem C. The map $\text{Res}_{\mathcal{C}}$ is Σ_2^0 -measurable if and only if $\mathcal{RC}$ is a quasivariety.

Theorem D. For the following quasivarieties, $\text{Res}_{\mathcal{C}}$ is Σ_2^0 -complete:

- the set of finitely generated k -solvable groups, for $k > 2$,
- the set of finitely generated exponent k groups, for k a big enough odd number,
- the set of finitely generated LEF groups,
- the set of finitely generated torsion free groups,
- the set of finitely generated left orderable groups.

Let $\text{Lim}_{\mathbb{N}} : \subseteq \mathbb{N}^{\mathbb{N}} \rightarrow \mathbb{N}$ be the function that maps a converging sequence of natural numbers to its limit.

Theorem E. The Weihrauch reduction

$$\text{Res}_{\mathcal{C}} \geq_W \text{Lim}_{\mathbb{N}}$$

holds if and only if $\mathcal{C}$ is not equationally Noetherian.

For each countable ordinal α , the function $\text{Lim}_{\searrow \alpha}$ maps a weakly descending sequence in α to its limit. See Section 4 for a precise definition.

Theorem F. We get the following Weihrauch reductions:

- For $\mathcal{C}$ the set of metabelian groups:

$$\text{Res}_{\mathcal{C}} \geq_W \text{Lim}_{\searrow \omega^\omega}.$$

- For $\mathcal{C}$ the set of p -nilpotent groups, for any $p \geq 1$,

$$\text{Res}_{\mathcal{C}} \equiv_W \text{Lim}_{\searrow \omega^2}.$$

- For $\mathcal{C}$ a finite set of finite groups,

$$\text{Res}_{\mathcal{C}} \equiv_W \text{Lim}_{\searrow \omega}.$$

The ordinals that appear in the above proof are related to the ordinal rank of the lattice of $\mathcal{C}$ marked groups, when $\mathcal{C}$ is an equationally Noetherian class. For a similar context in which these ordinals were found relevant, see [HRM26].

Theorem G. For $\mathcal{C}$ any of the sets of finite groups, of nilpotent groups or of finitely presented groups, $\text{Res}_{\mathcal{C}}$ is Σ_3^0 -measurable complete.

We also obtain some results in classical descriptive set theory. The following answers a problem of [BK22].

Theorem H. The set of residually finite groups is Σ_3^0 -complete. The same goes for the sets of residually nilpotent and residually finitely presented groups.

We summarize our results in Figure 1.1.

Notations. Let $\mathcal{C}$ be a class of groups.

$\text{Res}_{\mathcal{C}}$ denotes the function that maps a marked group to its greatest residually $\mathcal{C}$ quotient. Without further precision, we will consider that both the domain and co-domain of $\text{Res}_{\mathcal{C}}$ are equipped with the topology of the space of marked groups.

$\mathcal{RC}$ is the set of residually- $\mathcal{C}$ marked groups.

$\leq_W$ denotes continuous Weihrauch reduction.

$\leq$ denotes continuous translation between representations.

$\mathcal{G}_{\text{WP}}$ denotes the space of marked groups, which is associated to the representation associated to groups given by their word-problem.

$\mathcal{G}_{\text{Pres}}$ denotes the set of marked groups equipped with the Scott topology, and with the corresponding representation associated to presentations of groups.

By adding an exponent k to either one of the above, $\mathcal{G}_{\text{WP}}^k$ or $\mathcal{G}_{\text{Pres}}^k$, we indicate that we restrict our attention to k -marked groups.

2. GROUP THEORETICAL BACKGROUND

2.1. Marked groups. Fix $k \in \mathbb{N}$. A k -marked group is a finitely generated group together with a k -tuple of elements that generate it.

A morphism of k -marked groups from $(G, (s_1, \dots, s_k))$ to $(H, (s'_1, \dots, s'_k))$ is a group morphism ϕ between G and H which satisfies that $\phi(s_i) = s'_i$ for $i = 1..k$. Such a morphism is an isomorphism if the underlying group morphism is a group isomorphism, and marked groups are considered up to isomorphism.

Note that there is at most one morphism from a k -marked group to another k -marked group, and that if there are morphisms $(G, S) \rightarrow (H, S')$ and $(H, S') \rightarrow$

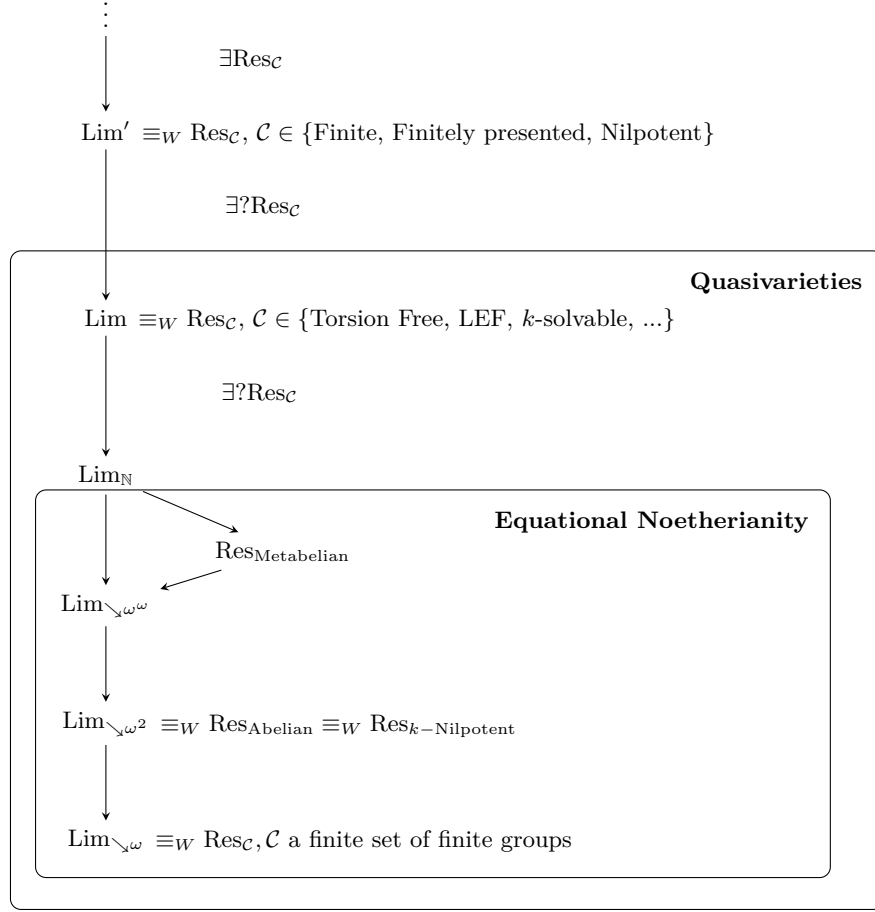


FIGURE 1.1. Summary

(G, S) , then (G, S) and (H, S') are isomorphic as marked groups: isomorphism classes of marked groups form a partially ordered set for the quotient relation.

Let $(\mathcal{G}_k, \rightarrow)$ be the poset of isomorphism classes of k -marked groups.

Remark that any isomorphism class of marked group can be identified with a normal subgroup of a rank k free group $\mathbb{F}_k$, by associating to a marked group (G, S) the kernel of the morphism $\mathbb{F}_k \rightarrow G$ which maps a basis of $\mathbb{F}_k$ to S . It follows from this that the poset $(\mathcal{G}_k, \rightarrow)$ is isomorphic to the lattice of normal subgroups of $\mathbb{F}_k$ under reverse inclusion.

Let $(\mathcal{G}, \rightarrow)$ be the poset of isomorphism classes of marked groups, obtained by taking the disjoint union of the sets $(\mathcal{G}_k, \rightarrow)$.

2.2. Quasivarieties. A (group) **quasi-identity** is a sentence of the form

$$(2.1) \quad \bigwedge_{i=1}^n w_i = 1 \implies r = 1,$$

for $(w_1, \dots, w_n, r)$ a finite tuple of elements of a free group $\mathbb{F}_S$. A group G satisfies the quasi-identity defined by $(w_1, \dots, w_n, r) \in \mathbb{F}_S^{n+1}$ if for any substitution of the generators of $\mathbb{F}_S$ by elements of G the sentence (2.1) is satisfied.

A class of groups $\mathcal{QV}$ is called a **group quasivariety** if there exists a set L of quasi-identities such that $\mathcal{QV}$ is the class of all groups that satisfy the quasi-identities of L .

Equivalently [BS81], a group quasivariety is a class of groups stable by forming subgroups, direct products, and ultraproducts.

A **group variety** is a quasivariety that can be defined by a set of quasi-identities of the form $1 = 1 \implies r = 1$. Such quasi-identities are called **identities**. Equivalently, by Birkhoff's Variety Theorem [BS81], a variety is a class of groups stable by quotients, subgroups and Cartesian products.

Example 2.1. The following are varieties:

- abelian groups,
- k -nilpotent groups,
- k -solvable groups,
- exponent k -group.

The following are quasivarieties:

- torsion free groups,
- left and bi-orderable groups [How82],
- locally indicable groups [How82],
- LEF groups,
- groups that do not contain Thompson's Group F.

Example 2.2. The set of residually free groups is a quasivariety, because free groups are equationally noetherian (see Lemma 2.6). The limit groups are known to be those residually free groups which are *commutative transitive*: for all (a, b, c) with $b \neq 1$, if a and b commute and b and c commute, then a and c commute. This expression is close to being a quasi-identity, but not quite:

$$\forall a, b, c, ([a, b] = 1 \wedge [b, c] = 1) \implies (b = 1 \vee [a, c] = 1).$$

The set of limit groups is in fact not a quasivariety, as $\mathbb{F}_2 \times \mathbb{F}_2$ is not a limit group.

2.3. Equational Noetherianity. For n in $\mathbb{N}$, consider a tuple $(X_1, \dots, X_n)$ of unknowns, and let $\mathbb{F}_n$ be the free group on $(X_1, \dots, X_n)$.

An *equation* is an element $W \in \mathbb{F}_n$.

If $(g_1, \dots, g_n)$ are elements in a group G , let $W(g_1, \dots, g_n)$ be the element of G obtained by substituting in W each element X_i by the corresponding g_i . We say that $(g_1, \dots, g_n)$ is a solution to W if $W(g_1, \dots, g_n) = 1$.

A set $Sys \subseteq \mathbb{F}_n$ is called a *system of equations*. For a system Sys of equations and a set of groups $\mathcal{C}$, we define the *algebraic set* $V_{Sys}(\mathcal{C})$ associated to Sys as

$$V_{Sys}(\mathcal{C}) = \{(g_1, \dots, g_n) \in G \mid G \in \mathcal{C}, W(g_1, \dots, g_n) = 1 \forall W \in Sys\}.$$

Definition 2.3 ([BMR99]). A group G is *equationally noetherian* if for every n and every tuple $(X_1, \dots, X_n)$ of unknown, every system of equations with unknowns $(X_1, \dots, X_n)$ over G is equivalent to a finite subsystem:

$$\forall Sys \subseteq \mathbb{F}_n, \exists Sys_0 \subseteq Sys, Sys_0 \text{ is finite \& } V_{Sys_0}(\{G\}) = V_{Sys_0}(\{G\}).$$

The above definition was extended to families in [GH19].

Definition 2.4 ([GH19], Definition A). A set $\mathcal{C}$ of groups is *equationally noetherian* if for every n and every tuple $(X_1, \dots, X_n)$ of unknowns, every system of equations with unknowns $(X_1, \dots, X_n)$ over $\mathcal{C}$ is equivalent to a finite subsystem.

We will mostly use the following characterization of equational noetherianity, which appears in [Hou07] for equationally noetherian groups, and in [GH19] for equationally noetherian families:

Lemma 2.5 ([Hou07, GH19]). *The following are equivalent for a family $\mathcal{C}$ of groups:*

- *The family $\mathcal{C}$ is equationally noetherian,*
- *Every residually $\mathcal{C}$ marked group is finitely presentable as a residually $\mathcal{C}$ group: for every residually $\mathcal{C}$ marked group (G, S) , there is a finitely presented marked group (H, S') with*

$$(G, S) = \text{Res}_{\mathcal{C}}(H, S').$$

- *Sequences of marked quotients of residually $\mathcal{C}$ groups stabilize: for every sequence $((G_n, S_n))_{n \in \mathbb{N}}$ of residually $\mathcal{C}$ marked groups with $(G_n, S_n) \rightarrow (G_{n+1}, S_{n+1})$ for all n , there is some n_0 with $(G_{n_0}, S_{n_0}) = (G_{n_0+1}, S_{n_0+1})$.*

The following lemma is a straightforward generalization of [Hou07, Theorem 2.1(5)] to equationally noetherian families.

Lemma 2.6. *Let $\mathcal{C}$ be a set of finitely generated groups. If $\mathcal{C}$ is equationally noetherian, then $\mathcal{RC}$ is a quasivariety.*

It follows that, on equationally noetherian families, the operator $\mathcal{C} \mapsto \mathcal{RC}$, which maps a class $\mathcal{C}$ of groups to the class of residually $\mathcal{C}$ groups, agrees with the operator $\mathcal{C} \mapsto \text{QVar}\mathcal{C}$, which maps a class of groups to the quasivariety it generates.

Proof of Lemma 2.6. We have that $\mathcal{RC}$ is not a quasivariety if and only if there is an infinite set of relations $\{r_i, i \in \mathbb{N}\}$ which implies a relation $w = 1$ in all groups in $\mathcal{C}$, and yet no finite subset of $\{r_i, i \in \mathbb{N}\}$ implies $w = 1$ in all groups in $\mathcal{C}$. But in this case the sequence of morphisms

$$\text{Res}_{\mathcal{C}}\langle S \mid r_0 \rangle \rightarrow \text{Res}_{\mathcal{C}}\langle S \mid r_0, r_1 \rangle \rightarrow \dots$$

cannot stabilize. Indeed, if it stabilized after n steps, it would follow that $\text{Res}_{\mathcal{C}}\langle S \mid r_0, i \leq n \rangle$ already satisfies all the relations of $\{r_i, i \in \mathbb{N}\}$, and thus that $w = 1$ in this group, yet by hypothesis this is not the case. $\square$

3. REPRESENTATIONS AND THE WEIHRAUCH LATTICE

In the following, we mostly follow the surveys [BGP21, Sch21], further references and historical remarks can also be found there.

3.1. Admissible representations, continuous multi-functions. A *representation* of a set X is a partial surjection $\rho : \subseteq \mathbb{N}^{\mathbb{N}} \rightarrow X$.

If $\rho(u) = x$, then u is called a ρ -name of x .

The *final topology* of a representation $\rho : \subseteq \mathbb{N}^{\mathbb{N}} \rightarrow X$ is the topology on X given by

$$O \text{ open} \iff \rho^{-1}(O) \text{ is open in } \text{dom}(\rho),$$

where $\text{dom}(\rho)$ is equipped with the subset topology from Baire space.

A *multi-function* between sets X and Y is a relation $R \subseteq X \times Y$ which we treat as a function f mapping points of X to subsets of Y via the formula $f(x) = \{y \in Y \mid (x, y) \in R\}$. We put $\text{dom}(f) = \{x \in X \mid f(x) \neq \emptyset\}$, and consider that a multi-function f between X and Y is *total* if $\text{dom}(f) = X$, and that it is *partial* otherwise.

The fact that f is a total multi-function from X to Y is denoted via $f : X \rightrightarrows Y$, and we denote a partial multi-function by $f : \subseteq X \rightrightarrows Y$.

Definition 3.1. Let (X, ρ) and (Y, τ) be represented spaces, and let $f : \subseteq X \rightrightarrows Y$ be a partial multi-function. A *realizer of f with respect to ρ and τ* , or (ρ, τ) -*realizer of f* , is a function $F : \subseteq \mathbb{N}^{\mathbb{N}} \rightarrow \mathbb{N}^{\mathbb{N}}$ which satisfies the following:

$$\forall u \in \text{dom}(\rho) \cap \rho^{-1}(\text{dom}(f)), \tau(F(u)) \in f(\rho(u)).$$

Note that the realizer of a multi-function is a single valued function.

The following diagram illustrates the notion of realizer:

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ \rho \uparrow & & \uparrow \tau \\ \mathbb{N}^{\mathbb{N}} & \xrightarrow{F} & \mathbb{N}^{\mathbb{N}} \end{array}$$

Let ρ and τ be representations of a set X . Say that ρ (*continuously*) *translates* to τ if the identity on X has a continuous (ρ, τ) -realizer, i.e. if there is a continuous partial function $F : \subseteq \mathbb{N}^{\mathbb{N}} \rightarrow \mathbb{N}^{\mathbb{N}}$ defined at least on $\text{dom}(\rho)$ and such that for any u in $\text{dom}(\rho)$, if u is a ρ -name of a point x in X , then $F(u)$ is a τ -name of x .

We denote the fact that ρ translates to τ by $\rho \leq \tau$. If $\rho \leq \tau$ and $\tau \leq \rho$, then ρ and τ are called (*continuously*) *equivalent*. When studying the continuous Weihrauch degrees of functions between represented spaces, representations can be considered up to equivalence.

Definition 3.2 ([KW85, Sch02]). Let X be a topological space. A representation $\rho : \subseteq \mathbb{N}^{\mathbb{N}} \rightarrow X$ is called *admissible for X* if the following hold:

- (1) The representation ρ is continuous,
- (2) For every continuous representation $\tau : \subseteq \mathbb{N}^{\mathbb{N}} \rightarrow X$ we have $\tau \leq \rho$.

It is possible for a representation ρ of a set X to be admissible for a topology $\mathcal{T}$ different from its final topology. But this arises only if this topology $\mathcal{T}$ is not sequential while having as sequentialization the final topology of ρ [Sch02, Sch21], and in all cases that we will consider, represented spaces are systematically equipped with the final topologies of their representations.

An *admissible representation* is a representation admissible with respect to its final topology. If ρ is admissible for X , (X, ρ) is an *admissibly represented space*.

A space that admits an admissible representation is necessarily T_0 (non T_0 spaces can be handled by using *multi-representations* [Sch02]).

Lemma 3.3 ([KW85, Sch02]). Let (X, ρ) be a represented space and let (Y, τ) be an admissibly represented space. Let $f : X \rightarrow Y$ be a function. Then the following are equivalent:

- (1) The function f is continuous with respect to the final topologies of ρ and τ .
- (2) The function f admits a continuous realizer.

The above lemma permits the definition of a robust notion of continuity for multi-functions between topological spaces that admit admissible representations.

Definition 3.4. A multi-function $f : \subseteq X \rightrightarrows Y$ between topological spaces X and Y is called *continuous* if it admits a continuous realizer with respect to some (equiv. any) admissible representations on X and Y .

Lemma 3.3 implies that if f is a single valued function, the above definition does coincide with continuity. It does not seem to be possible to define the above notion of continuity for multi-functions without going through the category of represented spaces. This is discussed in [BH94].

It is in fact very easy to construct admissible representations of second countable spaces.

Definition 3.5 (Kreitz-Weihrauch, [KW85]). Let X be a T_0 second countable topological space and $(B_i)_{i \in \mathbb{N}}$ a basis for X . The *standard representation associated to $(B_i)_{i \in \mathbb{N}}$* is given by

$$\forall u \in \mathbb{N}^{\mathbb{N}}, \rho(u) = x \iff \text{Im}(u) = \{n \in \mathbb{N} \mid x \in B_n\}.$$

Each numbered basis of X yields a standard representation. However, the choice of a numbered basis is inconsequential:

Theorem 3.6 ([KW85]). *All standard representations of a second countable space are admissible and continuously equivalent. Every admissible representation of a second countable space is continuously equivalent to a standard representation.*

3.2. Some constructions of representations.

3.2.1. Representation of a product. If (X, ρ) and (Y, τ) are represented spaces, the set $X \times Y$ is naturally equipped with the representation $\rho \times \tau$, given by

$$(\rho \times \tau)((u_n)_{n \in \mathbb{N}}) = (\rho((u_{2n})_{n \in \mathbb{N}}), \tau((u_{2n+1})_{n \in \mathbb{N}})).$$

The domain of $\rho \times \tau$ is the set of sequences $(u_n)_{n \in \mathbb{N}}$ such that $(u_{2n})_{n \in \mathbb{N}} \in \text{dom}(\rho)$ and $(u_{2n+1})_{n \in \mathbb{N}} \in \text{dom}(\tau)$.

3.2.2. Representation of an infinite product. Let $(n, k) \mapsto \langle n, k \rangle$ be the standard Cantor pairing function defined by $\langle n, k \rangle := \frac{1}{2}(n + k + 1)(n + k) + k$.

We define a pairing function $\langle \cdot \rangle : (\mathbb{N}^{\mathbb{N}})^{\mathbb{N}} \rightarrow \mathbb{N}^{\mathbb{N}}$ by $\langle p_0, p_1, p_2, \dots \rangle \langle n, k \rangle := p_n(k)$ for all $p_i \in \mathbb{N}^{\mathbb{N}}$ and $n, k \in \mathbb{N}$.

If (X, ρ) is a represented space, we define a representation $\delta_{X^{\mathbb{N}}}$ of $X^{\mathbb{N}}$ by:

$$\delta_{X^{\mathbb{N}}}(\langle p_0, p_1, p_2, \dots \rangle) = (\rho(p_n))_{n \in \mathbb{N}}.$$

3.2.3. Sierpiński space. The Sierpiński space $\mathbb{S}$ is the set $\{0, 1\}$ equipped with the topology $\{\emptyset, \{1\}, \{0, 1\}\}$. It admits an admissible representation $c_{\mathbb{S}} : \mathbb{N}^{\mathbb{N}} \rightarrow \mathbb{S}$ given by:

$$c_{\mathbb{S}}(0^\omega) = 0,$$

$$c_{\mathbb{S}}(u) = 1 \text{ for any } u \neq 0^\omega.$$

3.2.4. Restrictions. If (X, ρ) is a represented space and $Y \subseteq X$, we let $\rho|_Y$ be the restriction of ρ to $\rho^{-1}(Y)$. If X is a second countable space and ρ is admissible, then the final topology of $\rho|_Y$ is the subset topology on Y and $\rho|_Y$ is also admissible.

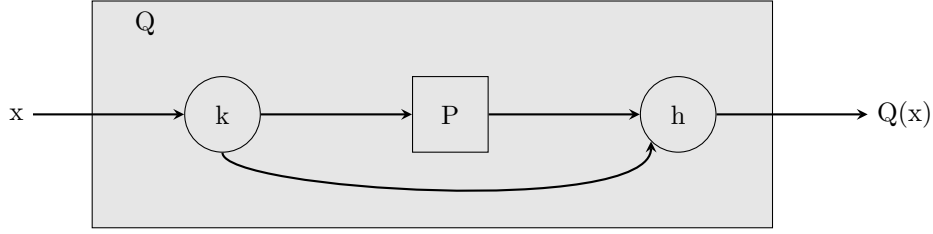


FIGURE 3.1. Weihrauch reducibility

3.2.5. *Jump of a represented space.* Let (X, δ) be a represented space. We define a new representation $\delta' : \subseteq \mathbb{N}^{\mathbb{N}} \rightarrow X$ of X by the following:

$$\delta'(p) = x \iff \begin{cases} p, \text{ seen as an element of } (\mathbb{N}^{\mathbb{N}})^{\mathbb{N}} \text{ via a pairing function,} \\ \text{is a sequence that converges to a } \delta\text{-name of } x \end{cases}$$

The represented space (X, δ') is called the *jump* of (X, δ) . Obviously $\delta \leq \delta'$. Note that the final topology of δ' is always the indiscrete topology.

3.3. Continuous Weihrauch reduction.

Definition 3.7. A multi-function between represented spaces is called a **problem**.

Let $P_1 : \subseteq X \rightrightarrows Y$ and $P_2 : \subseteq X \rightrightarrows Y$ be problems. We say that P_1 **solves** P_2 , (sometimes also: P_1 is a strengthening of P_2 and P_2 a weakening of P_1) and write $P_1 \sqsubseteq P_2$, if

$$\text{dom}(P_2) \subseteq \text{dom}(P_1) \ \& \ \forall x \in \text{dom}(P_2), P_1(x) \subseteq P_2(x).$$

When a multi-function P is interpreted as a problem, we see $P(x)$ as a set of valid outputs on input x . And thus if another problem Q satisfies $Q \sqsubseteq P$, it means that each valid output of Q is a valid output of P , hence the term “ Q solves P ”.

If $P : \subseteq X \rightrightarrows Y$ and $Q : \subseteq Z \rightrightarrows W$ are problems, their **product**, denoted by $P \times Q$, is the problem

$$\begin{aligned} P \times Q : \subseteq X \times Z &\rightrightarrows Y \times W \\ (u, v) &\mapsto P(u) \times Q(v). \end{aligned}$$

We can now define the topological version Weihrauch reduction.

Definition 3.8 (Continuous Weihrauch reduction). Let $P : \subseteq X \rightrightarrows Y$ and $Q : \subseteq Z \rightrightarrows W$ be problems. We say that P **(continuously) Weihrauch reduces** to Q , and write $P \geq_W Q$, if there is a represented space V and continuously realizable multi-functions $h : \subseteq V \times W \rightrightarrows Y$ and $k : \subseteq X \rightrightarrows V \times Z$ such that

$$h \circ (\text{id}_V \times P) \circ k \sqsubseteq Q.$$

3.4. **Comments on the definition of Weihrauch reducibility.** We want to illustrate the pros of allowing continuous multi-functions in the definition of Weihrauch reducibility in order to capture the idea of “a function being more discontinuous than another”.

Definition 3.9. We denote by LPO (for *Limited Principle of Omniscience*¹) the Weihrauch equivalence class of the following problem:

$$\text{LPO} : \mathbb{N}^{\mathbb{N}} \rightarrow \{0, 1\}, \text{LPO}(u) \mapsto \begin{cases} 1, & \text{if } u = 0^\omega, \\ 0, & \text{otherwise.} \end{cases}$$

LPO is the equivalence class of all problems whose discontinuity points are isolated, and it is the easiest discontinuous single valued problem. The following result was obtained by Weihrauch [Wei92] in the context of admissible representations of metric spaces, but it is easy to generalize it to T_0 second countable spaces.

Proposition 3.10. *A function $P : X \rightarrow Y$ between T_0 second countable spaces is discontinuous if and only if $P \geq_W \text{LPO}$.*

Example 3.11. The following problems are all instances of LPO, however, they would not be equivalent if we did not allow for multi-functions in the definition of Weihrauch reducibility:

- $f : \mathbb{R} \rightarrow \mathbb{R}, x \mapsto \begin{cases} 0, & \text{if } x < 1/3, \\ 1, & \text{otherwise;} \end{cases}$
- $\tilde{f}$, the restriction of f to the Cantor ternary set;
- the injection $P : \mathbb{S} \rightarrow \{0, 1\}$ of the Sierpiński space into $\{0, 1\}$ equipped with the discrete topology.

Indeed, any continuous function $\mathbb{S} \rightarrow \mathbb{R}$ is constant because $\mathbb{R}$ is T_2 , and similarly any continuous function from $\mathbb{R}$ to the Cantor ternary set is constant because $\mathbb{R}$ is connected. On the other hand, many non-constant continuous multi-functions exist between these spaces. For instance, if $(u_n)_{n \in \mathbb{N}}$ is a sequence of real numbers converging to $l \in \mathbb{R}$, then the multi-function $\mathbb{S} \rightrightarrows \mathbb{R}$ mapping the open point to $\{u_n, n \in \mathbb{N}\}$ and the closed point to l is continuous. We leave it to the reader to explicitly write down Weihrauch reductions between the above problems.

Here we give an example that illustrates the need to pass through the input.

Example 3.12. Consider the following functions:

- $f : \mathbb{R} \rightarrow \mathbb{R}, x \mapsto \begin{cases} 0, & \text{if } x < 0, \\ 1, & \text{otherwise;} \end{cases}$
- $\hat{f} : \mathbb{R} \rightarrow \mathbb{R}, x \mapsto \begin{cases} x, & \text{if } x < 0, \\ x + 1, & \text{otherwise.} \end{cases}$

One easily checks that both problems are equivalent to LPO. However, by using a definition of Weihrauch reducibility that does not include the id_V component in $h \circ (\text{id}_V \times P) \circ k \sqsubseteq Q$, those problems become non-equivalent.

The reduction obtained by removing the identity in the definition of Weihrauch reducibility is called strong Weihrauch reducibility [BGP21].

3.5. Some classical problems. We introduce some benchmark problems.

The problem *Limit on the natural numbers*, denoted by $\text{Lim}_{\mathbb{N}} : \subseteq \mathbb{N}^{\mathbb{N}} \rightarrow \mathbb{N}$, is defined on the subset of $\mathbb{N}^{\mathbb{N}}$ consisting of converging sequences, a sequence is mapped to its limit.

¹The name comes from constructive mathematics. The constructivist Errett Bishop was the first to talk about Omniscience principles.

The problem *Limit on Baire Space*, $\text{Lim} : \subseteq \mathbb{N}^{\mathbb{N}} \rightarrow \mathbb{N}^{\mathbb{N}}$ is defined thanks to a bijection between $\mathbb{N}^{\mathbb{N}}$ and $(\mathbb{N}^{\mathbb{N}})^{\mathbb{N}}$: Lim is defined on the set of elements of $\mathbb{N}^{\mathbb{N}}$ which, seen as elements of $(\mathbb{N}^{\mathbb{N}})^{\mathbb{N}}$, define a converging sequence, and then a sequence is mapped to its limit.

The *jump of a problem* $P : \subseteq X \rightrightarrows Y$ is the problem obtained by replacing the representation of X by its jump (Section 3.2.5). Thus extensionally it is the same multi-function, but the representation has been changed on the input side.

The n -th jump of a problem P is denoted by $P^{(n)}$.

A function $f : \subseteq \mathbb{N}^{\mathbb{N}} \rightarrow \mathbb{N}^{\mathbb{N}}$ is called Σ_n^0 -measurable if the preimage of any open set is in Σ_n^0 . A function between represented spaces is called Σ_n^0 -measurable if it admits a Σ_n^0 -measurable realizer.

A problem P is Σ_n^0 -measurable complete if for every problem Q that is Σ_n^0 -measurable we have $P \geq_W Q$.

Theorem 3.13 ([Bra04]). *Lim is Σ_2^0 -measurable complete, and more generally $\text{Lim}^{(n)}$ is Σ_{n+2}^0 -measurable complete.*

Example 3.14. The problem LPO' , the jump of LPO , is easily seen to be equivalent to the problem $\exists^\infty : \{0, 1\}^{\mathbb{N}} \rightarrow \{0, 1\}$, given by

$$\exists^\infty((u_n)_{n \in \mathbb{N}}) = 1 \iff \exists^\infty n, u_n = 1.$$

See for instance [PPS24].

3.6. Finite parallelizability. For problems $P : \subseteq X \rightrightarrows Y$ and $Q : \subseteq Z \rightrightarrows W$, recall that their product $P \times Q : \subseteq X \times Z \rightrightarrows Y \times W$ is given by

$$(P \times Q)(x, z) = P(x) \times Q(z),$$

with $\text{dom}(P \times Q) = \text{dom}(P) \times \text{dom}(Q)$.

The *finite parallelization* of a problem $P : \subseteq X \rightrightarrows Y$ is the problem

$$\begin{aligned} P^* : \subseteq \bigcup_{n \in \mathbb{N}} \{n\} \times X^n &\rightrightarrows \bigcup_{n \in \mathbb{N}} \{n\} \times Y^n \\ (i, x_1, \dots, x_i) &\mapsto \{i\} \times P(x_1) \times \dots \times P(x_i) \end{aligned}$$

with domain $\bigcup_{n \in \mathbb{N}} \{n\} \times \text{dom}(X)^n$.

In words: the instances of P^* consist in finitely many instances of P , and solutions for these instances are simply the tuples consisting of solutions for each instances.

A problem P is *finitely parallelizable* if $P \equiv_W P^*$. This is easily seen to be equivalent to the fact that $P \equiv_W P \times P$.

The problem Res_C is always finitely parallelizable (Proposition 7.2).

3.7. Infinite parallelization. Let $P : \subseteq X \rightrightarrows Y$ be a problem. The *infinite parallelization* of P is the problem $\hat{P} : \subseteq X^{\mathbb{N}} \rightrightarrows Y^{\mathbb{N}}$ given by

$$\hat{P}((u_n)_{n \in \mathbb{N}}) = (P(u_n))_{n \in \mathbb{N}}.$$

Here $X^{\mathbb{N}}$ and $Y^{\mathbb{N}}$ are equipped with the representations induced by those of X and Y as defined in Section 3.2.2.

We will use the following equalities [BGP21]:

Proposition 3.15.

$$\begin{aligned} \text{Lim} &\equiv_W \widehat{\text{LPO}}, \\ \text{Lim}' &\equiv_W \widehat{\text{LPO}'}. \end{aligned}$$

4. DISCRETIZATION PROBLEM ASSOCIATED TO THE ALEXANDROV TOPOLOGY OF AN ORDINAL

4.1. Rank function on a well-founded partial order. The following was used by Hertling [Her20] to study continuous Weihrauch degrees.

Definition 4.1. Let $(A, \leq)$ be a well founded order. We define a function $\text{rank}_{(A, \leq)} : A \rightarrow \text{ORD}$, where ORD is the class of ordinals, whose order we designate by $\leq$, inductively as follows:

$$\text{rank}_{(A, \leq)}(a) = \sup\{\text{rank}_{(A, \leq)}(b) + 1 \mid b < a\}$$

for any $a \in A$. (The base case is given by $\sup \emptyset = 0$.)

We define the rank of $(A, \leq)$ as

$$\text{rank}((A, \leq)) = \sup\{\text{rank}_{(A, \leq)}(a) + 1 \mid a \in A\}.$$

Example 4.2. The rank of an ordinal is that ordinal. The rank of the lattice of marked quotients of $(\mathbb{Z}^2, ((1, 0), (0, 1)))$ is $\omega 2 + 1$.

4.2. Alexandrov topology on a partially ordered set. If $(X, \leq)$ is a partially ordered set, we define a T_0 topology on it by

$$U \text{ is open} \iff U \text{ is an upper set,}$$

i.e. U is open iff $\forall x \in U, \forall y \in X, x \leq y \implies y \in U$. This defines *the Alexandrov topology* associated to $\leq$ [Are99]. This topology has the property that arbitrary intersections of open sets are open, and, in fact, any T_0 topology with the property that arbitrary intersections of open sets are open is the Alexandrov topology of its specialization order.

And we have the obvious properties:

Proposition 4.3. *The order preserving maps between partially ordered sets coincide with the continuous functions with respect to their associated Alexandrov topologies.* $\square$

Proposition 4.4. *The ordinal rank of a well-founded order $(A, \leq)$ is identical to the Cantor-Bendixson rank of the Alexandrov topology induced by $(A, \leq^{\text{op}})$.* $\square$

The following proposition will be useful to relate the ordinal rank to Weihrauch reductions.

A multi-function f is called injective if $\forall x, y \in \text{dom}(f), x \neq y \implies f(x) \cap f(y) = \emptyset$.

Proposition 4.5. *Let $(A, \leq)$ be a countable well founded order, and let α be its ordinal rank. Equip A and α with the Alexandrov topologies associated to the respective reverse orders $\leq^{\text{op}}$. Then the function $\text{rank}_{(A, \leq)} : A \rightarrow \alpha$ is continuous, and its multi-inverse $\text{rank}_{(A, \leq)}^{-1} : \alpha \rightrightarrows A$ is an injective continuous multi-function.*

Proof. The function $\text{rank}_{(A, \leq)} : A \rightarrow \alpha$ is order preserving, thus continuous. It is onto by definition of α .

Its multi-inverse $\text{rank}_{(A, \leq)}^{-1} : \alpha \rightrightarrows A$ is clearly injective.

We show that $\text{rank}_{(A, \leq)}^{-1}$ has a continuous realizer.

The Alexandrov topology on $(A, \leq^{\text{op}})$ is second countable, a countable basis being given by $(B_a)_{a \in A}$, where $B_a = \{b \in A \mid b \leq a\}$. It follows that the standard representation (Definition 3.5) associated to the Alexandrov topology on $(A, \leq^{\text{op}})$

is given by: a point a is represented by a weakly descending sequence in A which is eventually constant at a .

Consider a well ordering $\succeq$ on A (distinct from $\leq$ which need not be total).

Then we define a continuous realizer of $\text{rank}_{(A, \leq)}^{-1}$ by the following:

A weakly descending sequence $(\beta_n) \in \alpha^{\mathbb{N}}$ is mapped to a weakly descending sequence $(a_n) \in A^{\mathbb{N}}$ given by:

$$\begin{aligned} a_0 &= \inf_{\succeq} \{a \in A \mid \text{rank}_{(A, \leq)}(a) = \beta_0\} \\ a_{n+1} &= \inf_{\succeq} \{a \in A \mid \text{rank}_{(A, \leq)}(a) = \beta_{n+1} \text{ \& } a \leq a_n\}. \end{aligned}$$

This clearly defines a continuous realizer to $\text{rank}_{(A, \leq)}^{-1} : \alpha \rightrightarrows A$. $\square$

4.3. Discretization problem. The following notion will be useful. Let (X, ρ) be a countable represented space.

Lemma 4.6. *Let $f_1 : X \rightarrow \mathbb{N}$ and $f_2 : X \rightarrow \mathbb{N}$ be injections. Then $f_1 \equiv_W f_2$.*

Proof. Straightforward. $\square$

Definition 4.7. The *discretization problem of a countable represented space* (X, ρ) , denoted $\text{Dscr}((X, \rho))$, is the equivalence class of the problem $f : X \rightarrow \mathbb{N}$, for any injection f .

And:

Definition 4.8. The *discretization problem of a countable topological space* X , denoted $\text{Dscr}(X)$, is defined as $\text{Dscr}((X, \rho))$ for any admissible representation ρ of X .

4.4. Weakly descending sequence in an ordinal. Let α be a countable ordinal.

Fix a surjection $f : \mathbb{N} \rightarrow \alpha$. We consider the following problem:

$$\begin{aligned} \text{Lim}_{\searrow(\alpha, f)} &: \subseteq \mathbb{N}^{\mathbb{N}} \rightrightarrows \mathbb{N}, \\ \text{dom}(\text{Lim}_{\searrow(\alpha, f)}) &= \{(u_n)_{n \in \mathbb{N}} \in \mathbb{N}^{\mathbb{N}} \mid \forall n \in \mathbb{N}, f(u_{n+1}) \leq f(u_n)\}, \\ \forall (u_n)_{n \in \mathbb{N}} \in \text{dom}(\text{Lim}_{\searrow(\alpha, f)}), \text{Lim}_{\searrow(\alpha, f)}((u_n)_{n \in \mathbb{N}}) &= \{t \in \mathbb{N} \mid f(t) = \inf_{n \in \mathbb{N}} (f(u_n))\}. \end{aligned}$$

Thus the input to $\text{Lim}_{\searrow(\alpha, f)}$ is a weakly descending sequence in α , and the output is its limit. The above problem in fact does not depend on the chosen surjection f .

Lemma 4.9. *If f and g are two surjections $f, g : \mathbb{N} \rightarrow \alpha$, then $\text{Lim}_{\searrow(\alpha, f)} \equiv_W \text{Lim}_{\searrow(\alpha, g)}$.*

Proof. Left to the reader. $\square$

Definition 4.10. Define $\text{Lim}_{\searrow \alpha}$ to be the equivalence class of $\text{Lim}_{\searrow(\alpha, f)}$ for any surjection $f : \mathbb{N} \rightarrow \alpha$.

Lemma 4.11. *The map $\alpha \mapsto \text{Lim}_{\searrow \alpha}$ is an order preserving injection from the set of countable ordinals to the continuous Weihrauch degrees.*

Proof. That $\alpha \mapsto \text{Lim}_{\searrow \alpha}$ is order preserving is straightforward. Injectivity follows directly from Hertling's notion of level of discontinuity of a problem [Her96], which is an ordinal invariant similar to the Cantor-Bendixson rank for functions, and from the obvious fact that $\text{Lim}_{\searrow \alpha}$ has level α . $\square$

Example 4.12. The problem $\text{Lim}_{\searrow 2}$ is equivalent to LPO. The problem $\text{Lim}_{\searrow \omega}$ is LPO^* , the finite parallelization of LPO.

Notice the following easy proposition:

Proposition 4.13. *Let α be a countable limit ordinal. Then for any problem P , $P \geq_W \text{Lim}_{\searrow \alpha}$ if and only if $P \geq_W \text{Lim}_{\searrow \beta}$ for every $\beta < \alpha$.*

Proof. The direct implication is straightforward. For the converse, notice that an instance of $\text{Lim}_{\searrow \alpha}$ is a weakly descending sequence whose first term necessarily belongs to some $\beta < \alpha$. Fix the functions k_β and h_β that witness the reduction $P \geq_W \text{Lim}_{\searrow \beta}$ for each ordinal $\beta \in \alpha$. The total reduction for $P \geq_W \text{Lim}_{\searrow \alpha}$ is then: given (u_n) , a weakly descending sequence in α , let β be $u_0 + 1$, apply the reduction k_β followed by P followed by h_β , this gives the limit of the sequence (u_n) . $\square$

4.5. Discretization problems and limit in a countable ordinal. The problems $\text{Lim}_{\searrow \alpha}$ are related to discretization problems:

Proposition 4.14. *The problem $\text{Lim}_{\searrow \alpha}$ is the discretization problem associated to the Alexandrov topology on α^{op} .*

Proof. It suffices to note that the description of an element of α by a weakly descending sequence in α is exactly the standard Kreitz-Weihrauch representation associated to the Alexandrov topology on α^{op} (see Definition 3.5). $\square$

Note also:

Proposition 4.15. *Let (X, ρ) be a countable represented space without condensation, and let α be its Cantor-Bendixson rank. Then $\text{Dscr}((X, \rho)) \leq_W \text{Lim}_{\searrow \alpha}$.*

Proof. Straightforward. $\square$

5. DIFFERENT REPRESENTATIONS OF MARKED GROUPS

We introduce different representations and their associated final topologies.

5.1. Space of marked groups. As explained in Section 2.1, a k -marked group can be seen as a normal subgroup of $\mathbb{F}_k$. For each k , we fix a bijection $\theta_k : \mathbb{N} \rightarrow \mathbb{F}_k$. We can then identify the power set of $\mathbb{F}_k$ with $\{0, 1\}^{\mathbb{N}}$. The set of k -marked groups can thus be seen as a subset of $\{0, 1\}^{\mathbb{N}}$. By equipping $\{0, 1\}^{\mathbb{N}}$ with the Cantor space topology, the set of k -marked groups inherits the subset topology which makes it a compact Polish space, called *the space of k -marked groups*. We define *the space of marked groups* by taking the disjoint union topology of the spaces of k -marked groups.

This topology is associated to the admissible representation ρ_{WP} , defined by:

$$\begin{aligned} \text{dom}(\rho_{\text{WP}}) &= \{(u_n)_{n \in \mathbb{N}} \in \{0, 1\}^{\mathbb{N}} \mid \{\theta_{u_0}(i) \mid u_i = 1\} \text{ is a normal subgroup of } \mathbb{F}_{u_0}\}; \\ \rho_{\text{WP}}((u_n)_{n \in \mathbb{N}}) &= \mathbb{F}_{u_0} / \{\theta_{u_0}(i) \mid u_i = 1\}. \end{aligned}$$

In other words: the name of a marked group encodes its word problem. Hence the subscript WP: ρ_{WP} is called the *word problem representation*.

We denote by $\mathcal{G}_{\text{WP}}$ the space of marked groups, whose topology is associated to the admissible representation ρ_{WP} .

5.2. Scott topology. Each k -marked group can be seen as an element of $\{0, 1\}^{\mathbb{N}}$. If, instead of considering the product topology associated to the discrete topology on $\{0, 1\}$, we take the product topology associated to the Sierpiński topology on $\{0, 1\}$, with $\{1\}$ as only non-trivial open set, we obtain what is known as *the Scott topology* on $\{0, 1\}^{\mathbb{N}}$. We denote by $\mathbb{S}^{\mathbb{N}}$ the set of binary sequences equipped with the Scott topology.

A basis for this topology is given by sets of the form $\Omega_A = \{(u_i)_{i \in \mathbb{N}} \in \{0, 1\}^{\mathbb{N}} \mid \forall i \in A, u_i = 1\}$ for A a finite subset of $\mathbb{N}$. In terms of marked groups, the basic open sets have the forms “those marked groups that satisfy the relations $\{r_1, \dots, r_n\}$ ”, for any finite set of relations $\{r_1, \dots, r_n\}$.

The Scott topology is the one that appears naturally when we consider the representation where groups are given by presentations. We now give three equivalent ways of defining this representation.

Again a sequence of free groups equipped with bases $(\mathbb{F}_k, S_k)$ is fixed, and $\theta_k : \mathbb{N} \rightarrow \mathbb{F}_k$ is the short-lex bijection induced by the basis S_k .

The *presentation representation* ρ_{Pres} is given by

$$\begin{aligned} \rho_{\text{Pres}} : \mathbb{N}^{\mathbb{N}} &\rightarrow \mathcal{G} \\ (u_n)_{n \in \mathbb{N}} &\mapsto \langle S_{u_0} \mid \theta_{u_0}(u_i), i \geq 1 \rangle. \end{aligned}$$

The *full presentation representation* ρ_{FullPres} only uses maximal presentations, i.e. presentations where all the relations of a marked groups are listed. It is simply the restriction of the above representation to sequences that define a normal subgroup:

$$\text{dom}(\rho_{\text{FullPres}}) = \{(u_n)_{n \in \mathbb{N}} \in \mathbb{N}^{\mathbb{N}} \mid \{\theta_{u_0}(u_i), i \geq 1\} \text{ is a normal subgroup of } \mathbb{F}_{u_0}\};$$

$$\begin{aligned} \rho_{\text{FullPres}} : \subseteq \mathbb{N}^{\mathbb{N}} &\rightarrow \mathcal{G} \\ (u_n)_{n \in \mathbb{N}} &\mapsto \langle S_{u_0} \mid \theta_{u_0}(u_i), i \geq 1 \rangle. \end{aligned}$$

The *Sierpiński word-problem representation* ρ_{SWP} is similar to ρ_{WP} except that $\{0, 1\}$ is equipped with the Sierpiński topology: the set $\mathcal{G}^k$ of k -marked groups is seen as a subset of $\mathbb{S}^{\mathbb{N}}$ and equipped with the induced topology and representation. A formal definition is as follows. Let $c_{\mathbb{S}^{\mathbb{N}}}$ be the admissible representation of $\mathbb{S}^{\mathbb{N}}$, defined as the product representation of $c_{\mathbb{S}}$ (see Section 3.2). We define ρ_{SWP}^k to be the restriction of $c_{\mathbb{S}^{\mathbb{N}}}$ to sequences that define a marked group.

We then take ρ_{SWP} as the disjoint union representation:

$$\begin{aligned} \text{dom}(\rho_{\text{SWP}}) &= \{(u_n)_{n \in \mathbb{N}} \in \mathbb{N}^{\mathbb{N}} \mid (u_n)_{n \geq 1} \in \text{dom}(\rho_{\text{SWP}}^{u_0})\}; \\ \forall (u_n)_{n \in \mathbb{N}} \in \text{dom}(\rho_{\text{SWP}}), \rho_{\text{SWP}}((u_n)_{n \in \mathbb{N}}) &= \rho_{\text{SWP}}^{u_0}((u_n)_{n \geq 1}). \end{aligned}$$

By Lemma 6.1 and Corollary 6.2 proved below, $\rho_{\text{Pres}} \equiv \rho_{\text{FullPres}} \equiv \rho_{\text{SWP}}$. By the following lemma, these three representations are admissible.

Lemma 5.1. *The representation ρ_{FullPres} is admissible for the Scott topology on $\mathcal{G}$.*

Proof. The Kreitz-Weihrach standard representation associated to the natural basis for the Scott topology on $\mathcal{G}$ is the following: the name of a group is a list that contains all finite sets of relations that this group satisfies. But this is clearly equivalent to the representation ρ_{FullPres} . $\square$

We denote by $\mathcal{G}_{\text{Pres}}$ the set of marked groups equipped with the representation ρ_{Pres} whose final topology is the Scott topology on the lattice of marked groups.

6. MOVING BETWEEN REPRESENTATIONS

6.1. Different representations of $\mathcal{G}_{\text{Pres}}$.

Lemma 6.1. *The representations ρ_{Pres} and ρ_{SWP} are equivalent.*

Proof. We first show that $\rho_{\text{Pres}} \leq \rho_{\text{SWP}}$.

Recall that for each $k > 0$, $\theta_k : \mathbb{N} \rightarrow \mathbb{F}_k$ is a bijection between $\mathbb{N}$ and a fixed rank k free group.

Given a presentation $\langle s_0, \dots, s_k \mid r_0, r_1, r_2, \dots \rangle$ for a marked group (G, S) , we define a double sequence $(t_{i,j})_{(i,j) \in \mathbb{N}^2} \in \{0, 1\}^{\mathbb{N}^2}$ by the following: $t_{i,j} = 1$ if $\theta_k(i)$ can be written as a product of at most j conjugates of the elements $\{r_0, \dots, r_j\}$, the conjugators having themselves length at most j . Otherwise $t_{i,j} = 0$.

It is clear that $(t_{i,j})_{(i,j) \in \mathbb{N}^2}$ depends continuously on the presentation $\langle s_0, \dots, s_k \mid r_0, r_1, r_2, \dots \rangle$. Furthermore, $\theta_k(i) = 1$ in (G, S) if and only if there exists some j such that $t_{i,j} = 1$, and otherwise $t_{i,j} = 0$ for all $j \in \mathbb{N}$. Thus $(t_{i,j})_{(i,j) \in \mathbb{N}^2}$ is a ρ_{SWP} -name of (G, S) .

We now show that $\rho_{\text{SWP}} \leq \rho_{\text{Pres}}$.

Consider a marked group (G, S) given by a double sequence $(t_{i,j})_{(i,j) \in \mathbb{N}^2} \in \{0, 1\}^{\mathbb{N}^2}$ that satisfies $\theta_k(i) = 1$ in (G, S) if and only if $\exists j \in \mathbb{N}, t_{i,j} = 1$. Consider Cantor's pairing function $(i, j) \mapsto \langle i, j \rangle$. We define a presentation $\langle s_0, \dots, s_k \mid r_0, r_1, r_2, \dots \rangle$ for (G, S) by $r_{\langle i, j \rangle} = \theta_k(i)$ if $t_{i,j} = 1$, otherwise $r_{\langle i, j \rangle} = 1_{\mathbb{F}_k}$. It is clear that this indeed defines a presentation of (G, S) , and the sequence $(r_i)_{i \in \mathbb{N}}$ depends continuously on $(t_{i,j})_{(i,j) \in \mathbb{N}^2}$. $\square$

Corollary 6.2. *The representations ρ_{Pres} and ρ_{FullPres} are equivalent.*

Proof. The translation $\rho_{\text{FullPres}} \leq \rho_{\text{Pres}}$ is immediate. And $\rho_{\text{Pres}} \leq \rho_{\text{FullPres}}$ follows from the proof given above of $\rho_{\text{SWP}} \leq \rho_{\text{Pres}}$: the constructed translation is easily seen to give a ρ_{FullPres} -name of the given marked group. $\square$

6.2. The identity from $\mathcal{G}_{\text{Pres}}$ to $\mathcal{G}_{\text{WP}}$. Here we study the complexity of $\text{Id} : \mathcal{G}_{\text{Pres}} \cap \mathcal{C} \rightarrow \mathcal{G}_{\text{WP}}$, i.e. of the identity function on a set $\mathcal{C}$ of groups, seen as going from the set of marked groups equipped with the Scott topology and the presentation representation to the space of marked groups equipped with the word problem representation.

Lemma 6.3. *We have $\text{Id} : \mathcal{G}_{\text{Pres}} \rightarrow \mathcal{G}_{\text{WP}} \equiv_W \text{Lim}$.*

Proof. The fact that $\text{Id} : \mathcal{G}_{\text{Pres}} \rightarrow \mathcal{G}_{\text{WP}} \leq_W \text{Lim}$ follows directly from the corresponding well known result for the Cantor space [BGP21]:

$$\text{Id} : \mathbb{S}^{\mathbb{N}} \rightarrow \{0, 1\}^{\mathbb{N}} \equiv_W \widehat{\text{LPO}} \equiv_W \text{Lim}.$$

Indeed, for each $k \in \mathbb{N}$, $\mathcal{G}_{\text{Pres}}^k$ is a subset of $\mathbb{S}^{\mathbb{N}}$ equipped with the induced topology, and $\mathcal{G}_{\text{WP}}^k$ is a subset of $\{0, 1\}^{\mathbb{N}}$ also equipped with the induced topology. And thus $\text{Id} : \mathcal{G}_{\text{Pres}} \rightarrow \mathcal{G}_{\text{WP}}$ is the restriction of a problem below Lim , it is again below Lim .

To prove that $\text{Id} : \mathcal{G}_{\text{Pres}} \rightarrow \mathcal{G}_{\text{WP}} \geq_W \text{Lim}$, we use an infinite set of independent relations. For $A \subseteq \mathbb{N}$, consider the marked group

$$(G_A, S) = \langle a, b, c, d \mid a^{-i}ba^i = c^{-i}dc^i, i \in A \rangle.$$

Using the fact that G_A is defined as an amalgamated product, it is easy to see that $\Psi : A \mapsto (G_A, S)$ is injective, and that:

- (1) The map $\Psi_1 : \mathbb{S}^{\mathbb{N}} \rightarrow \mathcal{G}_{\text{Pres}}$ is a homeomorphism onto its image,

(2) The map $\Psi_2 : \{0, 1\}^{\mathbb{N}} \rightarrow \mathcal{G}_{\text{WP}}$ is a homeomorphism onto its image.

It follows from this that $\text{Id} : \mathbb{S}^{\mathbb{N}} \rightarrow \{0, 1\}^{\mathbb{N}}$ can be written as $\Psi_2^{-1} \circ (\text{Id} : \mathcal{G}_{\text{Pres}} \rightarrow \mathcal{G}_{\text{WP}}) \circ \Psi_1$, this gives the desired reduction for $\text{Lim} \leq_W \text{Id} : \mathcal{G}_{\text{Pres}} \rightarrow \mathcal{G}_{\text{WP}}$. $\square$

The following lemma is proved exactly as above, we are simply explicitly asking for the existence of a function Ψ such as appeared in the above proof.

Lemma 6.4. *Let $\mathcal{C}$ be a set of finitely generated groups. Suppose that there is an injective function $\Psi : \{0, 1\}^{\mathbb{N}} \rightarrow \mathcal{G} \cap \mathcal{C}$, and that*

- (1) *The function $\Psi : \mathbb{S}^{\mathbb{N}} \rightarrow \mathcal{G}_{\text{Pres}} \cap \mathcal{C}$ is continuous,*
- (2) *The function $\Psi^{-1} : \mathcal{G}_{\text{WP}} \cap \text{Im}(\Psi) \rightarrow \{0, 1\}^{\mathbb{N}}$ is also continuous.*

Then $\text{Id} : \mathcal{G}_{\text{Pres}} \cap \mathcal{C} \rightarrow \mathcal{G}_{\text{WP}} \equiv_W \text{Lim}$.

Note that the existence of a function Ψ as above will always rely on the existence of an *infinitely independently presented group relative to $\mathcal{C}$* . We introduce this notion here. A classical notion is:

Definition 6.5 ([BCGS14]). Let G be a group and $\{N_i \mid i \in I\}$ a family of normal subgroups of G . This family is called *independent* if no subgroup is contained in the subgroup generated by the remaining groups:

$$\forall i_0 \in I, N_{i_0} \not\subseteq \langle N_i, i \neq i_0 \rangle.$$

We say that a finitely generated group G is *infinitely independently presentable* (or *INIP*) if there is a morphism $\pi : \mathbb{F}_k \twoheadrightarrow G$ from a free group of finite rank onto G whose kernel is generated by an infinite and independent family of normal subgroups.

We consider a relative version of the above.

Definition 6.6. Let $\mathcal{C}$ be a set of finitely generated groups. Let G be a group and $\{N_i \mid i \in I\}$ a family of normal subgroups of G . This family is called *independent relative to $\mathcal{C}$* if for every i_0 in I , there is a quotient $\psi_{i_0} : G \rightarrow H$ onto a group $H \in \mathcal{C}$ such that

$$\psi_{i_0}(N_{i_0}) \not\subseteq \langle \psi_{i_0}(N_i), i \neq i_0 \rangle.$$

We say that a finitely generated group G is *infinitely independently presentable relative to $\mathcal{C}$* (or *INIP relative to $\mathcal{C}$*) if there is a morphism $\pi : \mathbb{F}_k \twoheadrightarrow G$ from a free group of finite rank onto G whose kernel is generated by an infinite family of normal subgroups which is independent relative to $\mathcal{C}$.

Proposition 6.7. *Let $\mathcal{C}$ be a set of finitely generated groups. Suppose that there exists a map $\Psi : \mathcal{P}(\mathbb{N}) \rightarrow \mathcal{G} \cap \mathcal{C}$ that satisfies the conditions of Lemma 6.4. Then $\Psi(\emptyset)$ is infinitely independently presentable relative to $\mathcal{C}$.*

Proof. One needs to remark that, because Scott open sets are upward closed, the map Ψ must be order preserving: if $A \subseteq B$, then there is a marked group morphism $\Psi(A) \rightarrow \Psi(B)$. Consider the morphism $\pi : \mathbb{F}_k \rightarrow \Psi(\emptyset)$ that defines the marking of $\Psi(\emptyset)$. The family $\{N_i \mid i \in \mathbb{N}\}$ given by $N_i = \pi^{-1}(\ker(\Psi(\emptyset) \rightarrow \Psi(\{i\})))$ is the desired set of normal subgroups independent relative to $\mathcal{C}$. $\square$

Proposition 6.8. *Let $\mathcal{C}$ be a set of finitely generated groups stable by finite direct products. The problem $\text{Id} : \mathcal{G}_{\text{Pres}} \cap \mathcal{C} \rightarrow \mathcal{G}_{\text{WP}}$ is finitely parallelizable.*

Proof. Indeed, having a ρ_{WP} name for (G, S) and one for (H, S') is equivalent to having one for $(G \times H, S \cup S')$. $\square$

Proposition 6.9. *Let $\mathcal{C}$ be a set of finitely generated groups, such that every group in $\mathcal{C}$ admits a finite presentation relative to $\mathcal{C}$. Then the Alexandrov and Scott topologies coincide on the set of marked groups in $\mathcal{C}$.*

Proof. The Alexandrov topology is always finer than the Scott topology. Thus all we have to show is that any open in the Alexandrov topology is also Scott open. But a basis of the Alexandrov topology is given by sets of the form “all marked quotients of (G, S) ”. But if (G, S) admits a finite presentation $\langle S \mid R \rangle$ relative to $\mathcal{C}$, this set is identical to “all marked groups that satisfy the relations of R ”, and this is a Scott open set. $\square$

Proposition 6.10. *Suppose that $\mathcal{C}$ is equationally noetherian. Let $k \in \mathbb{N}$. Let α be the rank of the lattice of residually- $\mathcal{C}$ k -marked groups. Then*

$$\text{Id} : \mathcal{G}_{\text{Pres}}^k \cap \mathcal{RC} \rightarrow \mathcal{G}_{\text{WP}}^k \leq_W \text{Lim}_{\searrow \alpha}.$$

Proof. This follows from directly from Proposition 4.4, Proposition 6.9 and Proposition 4.15. $\square$

We now show that when the rank of the lattice of residually- $\mathcal{C}$ k -marked groups is below ω^ω , the previous bound is actually sharp.

Theorem 6.11. *Suppose that $\mathcal{C}$ is equationally noetherian. Let $k \in \mathbb{N}$. Let α be the rank of the lattice of residually- $\mathcal{C}$ k -marked groups. Suppose that $\alpha < \omega^\omega$. Then*

$$\text{Id} : \mathcal{G}_{\text{Pres}}^k \cap \mathcal{RC} \rightarrow \mathcal{G}_{\text{WP}}^k \equiv_W \text{Lim}_{\searrow \alpha}.$$

Proof. For conciseness, we denote the problem $\text{Id} : \mathcal{G}_{\text{Pres}}^k \cap \mathcal{RC} \rightarrow \mathcal{G}_{\text{WP}}^k$ by P .

Consider a multi-injection e of α into the lattice of residually- $\mathcal{C}$ k -marked groups, as given by Proposition 4.5.

We will in fact not only use this multi-injection e , but also its continuous realizer given in the proof of Proposition 4.5: fix a well order $\succeq$ on $\mathcal{G}_{\text{Pres}}^k \cap \mathcal{RC}$, the realizer of e is then defined by mapping a weakly descending sequence $(\beta_n) \in \alpha^\mathbb{N}$ to the sequence $((G_n, S_n)) \in \mathcal{RC}^\mathbb{N}$ given by

$$(G_{n+1}, S_{n+1}) = \inf_{\succeq} \left\{ (G, S) \mid \text{rank}_{\mathcal{G}_{\text{Pres}}^k}((G, S)) = \beta_{n+1} \ \& \ (G_n, S_n) \rightarrow (G, S) \right\}.$$

This descending sequence of marked groups can be seen as a single group presentation by putting together all the relations of the marked groups in $\{(G_n, S_n)\}$.

Denote by E this continuous realizer of e .

We will prove that for all natural number n :

- (1) If $P \geq_W \text{Lim}_{\searrow \omega^n}$ and $\alpha > \omega^{n+1}$, then $P \geq \text{Lim}_{\searrow \omega^{n+1}}$.
- (2) If $P \geq_W \text{Lim}_{\searrow \omega^n}$ and $\omega^{n+1} \geq \alpha$, then $P \geq \text{Lim}_{\searrow \alpha}$.

Note that $P \geq_W \text{Lim}_{\searrow \omega}$ is automatic because P is finitely parallelizable (by Proposition 6.8, noting that $\mathcal{RC}$ is stable by direct products) and discontinuous, and thus (1) and (2) together imply that $P \geq \text{Lim}_{\searrow \alpha}$.

We first prove (1). Suppose that $P \geq_W \text{Lim}_{\searrow \omega^n}$. Note that $\omega^{n+1} = \omega \omega^n$, and the elements of ω^{n+1} can be seen as pairs (u, v) with $u \in \omega$ and $v \in \omega^n$, the order being lexicographic with priority given to the right hand side.

We thus define maps $\pi_1 : \omega^{n+1} \rightarrow \omega$, $\delta = (u, v) \mapsto u$ and $\pi_2 : \omega^{n+1} \rightarrow \omega^n$, $\delta = (u, v) \mapsto v$. The map π_2 is order preserving.

Applying E to a name u of some $\delta \in \omega^{n+1}$ yields a group presentation $E(u)$ of a marked group with ordinal rank δ .

The problem P can be used to convert this ρ_{Pres} -name into a ρ_{WP} -name for $E(u)$.

Also, by hypothesis, the problem P can be used to obtain a name of $\pi_2(\delta)$ for the discrete topology.

And because P is finitely parallelizable, one application of P is sufficient to get both results.

We now prove that δ can be recovered from this data.

Since we have $\pi_2(\delta)$, it suffices to compute $\pi_1(\delta)$.

An upper bound to $\pi_1(\delta)$ can be computed from $\pi_2(\delta)$, simply by waiting for the weakly descending sequence that describes δ to reach a point of the form $(v, \pi_2(\delta))$, at which point we know that $\pi_1(\delta) \in \{0, \dots, v\}$.

Once this is known, there are finitely many possibilities for what the group $E(u)$ is. Indeed, $E(u)$ is either the marked group with ordinal rank $(v, \pi_2(\delta))$ that E produces upon the beginning of the name of δ , or its first (with respect to the well order $\succeq$) quotient with ordinal rank $(v-1, \pi_2(\delta))$, or its first quotient with ordinal rank $(v-2, \pi_2(\delta))$, ..., and so on until $(0, \pi_2(\delta))$.

But having access to the word problem of this marked group provides a sufficient amount of information to determine which alternative holds, since the topology of the space of marked groups induces the discrete topology on any finite set of marked groups.

We now prove (2), the proof is almost identical to that of (1).

Suppose that $P \geq_W \text{Lim}_{\omega^n}$ and $\omega^{n+1} \geq \alpha$. We again identify ω^{n+1} with $\omega \times \omega^n$ via the projection (π_1, π_2) . Given $\delta \in \alpha$ and a name u of it, we can use P to compute a ρ_{WP} -name of $E(u)$ and a name of $\pi_2(\delta)$ for the discrete topology. We conclude as for (1). $\square$

Notice that we have also proved:

Theorem 6.12. *Suppose that $\mathcal{C}$ is equationally noetherian. Let $k \in \mathbb{N}$. Let α be the rank of the lattice of residually- $\mathcal{C}$ k -marked groups. Suppose that $\alpha \geq \omega^\omega$. Then*

$$\text{Id} : \mathcal{G}_{\text{Pres}}^k \cap \mathcal{RC} \rightarrow \mathcal{G}_{\text{WP}}^k \geq_W \text{Lim}_{\omega^\omega}.$$

Proof. Following the proof of Theorem 6.11, we see that situation (2) never arises, and thus by using (1) we get that $\text{Id} : \mathcal{G}_{\text{Pres}}^k \cap \mathcal{RC} \rightarrow \mathcal{G}_{\text{WP}}^k \geq_W \text{Lim}_{\omega^n}$ for every n . We conclude by Lemma 4.13. $\square$

The following corollary is immediate.

Corollary 6.13. *Suppose that $\mathcal{C}$ is equationally noetherian. Let α_k be the ordinal rank of the lattice of residually- $\mathcal{C}$ k -marked groups, and let $\alpha = \sup(\alpha_k)$. If, for each k , $\alpha_k < \omega^\omega$, then*

$$\text{Id} : \mathcal{G}_{\text{Pres}} \cap \mathcal{RC} \rightarrow \mathcal{G}_{\text{WP}} \equiv_W \text{Lim}_{\omega^\omega}.$$

Proposition 6.14. *Suppose that $\mathcal{C}$ is not equationally noetherian. Then*

$$\text{Id} : \mathcal{G}_{\text{Pres}}^k \cap \mathcal{RC} \rightarrow \mathcal{G}_{\text{WP}}^k \geq_W \text{Lim}_{\mathbb{N}}.$$

Proof. By Lemma 2.5 there is a marked residually $\mathcal{C}$ group which does not admit a finite presentation as a residually $\mathcal{C}$ group. Let $\langle S \mid r_1, r_2, r_3, \dots \rangle$ be a presentation for this group. We can suppose w.l.o.g. that each marked quotient $\text{Res}_{\mathcal{C}} \langle S \mid r_1, \dots, r_n \rangle \rightarrow \text{Res}_{\mathcal{C}} \langle S \mid r_1, \dots, r_n, r_{n+1} \rangle$ is a strict quotient.

For each $n \in \mathbb{N}$ fix an infinite presentation $\pi_n = \langle S \mid t_1^n, t_2^n, t_3^n, \dots \rangle$ of $\text{Res}_C \langle S \mid r_1, \dots, r_n \rangle$.

Let $(u_n)_{n \in \mathbb{N}} \in \mathbb{N}^{\mathbb{N}}$ be a converging sequence. Define a presentation associated to $(u_n)_{n \in \mathbb{N}} \in \mathbb{N}^{\mathbb{N}}$ as follows. Define a function m_c by $m_c(n) = \#\{k \leq n \mid u_k \neq u_{k+1}\}$. Note that because $(u_n)_{n \in \mathbb{N}}$ converges, $m_c(n)$ converges to a certain limit l .

Define a presentation $\Pi = \langle S \mid v_1, v_2, v_3, \dots \rangle$ by

$$v_{\langle p, q \rangle} = t_q^{m_c(p)}.$$

It follows that Π defines the same marked group as π_l (with many redundant relators). Denote this marked group (G_Π, S) . Note that $(G_\Pi, S) \in \mathcal{RC}$.

Applying $\text{Id} : \mathcal{G}_{\text{Pres}}^k \cap \mathcal{RC} \rightarrow \mathcal{G}_{\text{WP}}^k$ to Π yields a solution to the word-problem in (G_Π, S) . From this solution, the least n_0 such that r_{n_0} is not a relation of (G_Π, S) can be computed.

It follows that the value of $(u_n)_{n \in \mathbb{N}}$ changes exactly n_0 times. This information is sufficient to recover the limit of $(u_n)_{n \in \mathbb{N}}$. $\square$

7. GENERAL RESULTS ON THE MAP Res_C

Proposition 7.1. *Let $\mathcal{C}$ be a set of finitely generated groups. The following are equivalent:*

- (1) *The set $\mathcal{RC}$ is a quasivariety;*
- (2) *The function $\text{Res}_C : \mathcal{G}_{\text{Pres}} \rightarrow \mathcal{G}_{\text{Pres}}$ is continuous;*
- (3) *The function $\text{Res}_C : \mathcal{G}_{\text{WP}} \rightarrow \mathcal{G}_{\text{Pres}}$ is continuous.*

Proof. (1) $\implies$ (2). Suppose that $\mathcal{RC}$ is a quasivariety. Given a presentation $\pi = \langle s_1, \dots, s_k \mid R \rangle$ for a marked group (G, S) , a presentation of $\text{Res}_C((G, S))$ is obtained by adding as further relators all elements w that appear in a quasi-identity $\bigwedge r_i = 1 \implies w = 1$ of $\mathcal{RC}$.

By Corollary 6.2, there is a continuous map F which maps any presentation of (G, S) to a full presentation of (G, S) , i.e. one where all relations of (G, S) appear as relators. There is also a continuous map H which maps a presentation of (G, S) to a presentation in which the “obvious consequences” of the quasi-identities of $\mathcal{RC}$ are added: for π a presentation, $H(\pi)$ contains the relations of π and all relations w which appear in the right hand side of a quasi-identity $\bigwedge_{1 \leq i \leq n} r_i = 1 \implies w = 1$ of $\mathcal{RC}$ for which $(r_1, \dots, r_n)$ are relations that appear in π . The composition of F and H yields the desired continuous realizer of $\text{Res}_C : \mathcal{G}_{\text{Pres}} \rightarrow \mathcal{G}_{\text{Pres}}$.

(2) $\implies$ (3) is immediate since the space of marked groups topology is finer than the Scott topology.

(3) $\implies$ (1). Suppose that $\mathcal{RC}$ is not a quasivariety. Then there exists $k \in \mathbb{N}$, an infinite set of relations $(r_1, r_2, \dots) \in \mathbb{F}_k^{\mathbb{N}}$, and a relation $w \in \mathbb{F}_k$ such that any marked group in $\mathcal{RC}$ which satisfies all the relations $(r_1, r_2, \dots)$ also satisfies $w = 1$, and yet for every finite subset A of $(r_1, r_2, \dots)$ there is a group in $\mathcal{RC}$ which satisfies A and $w \neq 1$. Consider the sequence $((G_n, S_n))_{n \in \mathbb{N}}$ of finitely presented groups given by $(G_n, S_n) = \langle s_1, \dots, s_k \mid r_1, \dots, r_n \rangle$. The sequence $((G_n, S_n))_{n \in \mathbb{N}}$ being increasing, it converges towards a marked group (G, S) . And yet, by construction, w is a relation in $\text{Res}_C((G, S))$, but it is never a relation of $\text{Res}_C((G_n, S_n))$. Thus $(\text{Res}_C((G_n, S_n)))_{n \in \mathbb{N}}$ does not converge to $\text{Res}_C((G, S))$. $\square$

Proposition 7.2. *Let $\mathcal{C}$ be a set of finitely generated groups. The map Res_C is finitely parallelizable.*

Proof. It suffices to show that $\text{Res}_{\mathcal{C}} \equiv_W \text{Res}_{\mathcal{C}} \times \text{Res}_{\mathcal{C}}$, which is obvious: computing the residually $\mathcal{C}$ image of (G, S) and that of (H, S') is equivalent to computing that of $(G \times H, S \cup S')$. $\square$

Proposition 7.3 (Arbitrarily complicated $\text{Res}_{\mathcal{C}}$). *There are $2^{2^{\aleph_0}}$ problems of the form $\text{Res}_{\mathcal{C}}$, and thus some are not Borel measurable.*

Proof. Consider a set $\mathcal{S}$ of finitely generated simple groups. Consider $\mathcal{C}_{\mathcal{S}} = \{G \mid \text{No group of } \mathcal{S} \text{ embeds in } G\}$.

We consider the map $\text{Res}_{\mathcal{C}_{\mathcal{S}}} : \mathcal{G}_{\text{WP}} \rightarrow \mathcal{G}_{\text{WP}}$.

The restriction of $\text{Res}_{\mathcal{C}_{\mathcal{S}}}$ to the (Polish) space of marked simple groups is Weihrauch equivalent to the characteristic function of $\mathcal{S}$ (given a simple group, is it in $\mathcal{S}$?), since non-trivial morphisms from simple groups are injective.

Since there are $2^{\aleph_0}$ finitely generated simple groups, there are $2^{2^{\aleph_0}}$ sets of simple groups, and thus $2^{2^{\aleph_0}}$ different problems $\text{Res}_{\mathcal{C}_{\mathcal{S}}}$. Notice that Weihrauch equivalence classes of problems defined on the space of marked groups have cardinality at most the continuum, and thus these $2^{2^{\aleph_0}}$ problems yield $2^{2^{\aleph_0}}$ Weihrauch degrees. $\square$

Proposition 7.4. *Let $\mathcal{C}$ be a countable set of finitely generated groups. Then $\text{Res}_{\mathcal{C}} : \mathcal{G}_{\text{WP}} \rightarrow \mathcal{G}_{\text{WP}} \leq_W \text{Lim}'$.*

Proof. By Theorem 3.13 it suffices to show that $\text{Res}_{\mathcal{C}}$ is Σ_3^0 -measurable.

Let $\mathcal{C} = \{(G_1, S_1), (G_2, S_2), \dots\}$ be an enumeration of all k -marked groups of $\mathcal{C}$.

Consider a basic clopen set Ω of the space of k -marked groups defined by two tuples of words $(w_1, \dots, w_n)$ and $(r_1, \dots, r_{n'})$: the set of k -marked groups which satisfy the relations $(r_1, \dots, r_{n'})$ and in which $(w_1, \dots, w_n)$ are not relations.

Decompose Ω as a finite intersection: groups in which $w \neq 1$, for each $w \in \{w_1, \dots, w_n\}$, and in which $r = 1$, for each $r \in \{r_1, \dots, r_{n'}\}$.

Notice that for each $n \in \mathbb{N}$, the set of marked groups that map onto (G_n, S_n) , $\{\Gamma \in \mathcal{G}_k \mid \Gamma \rightarrow (G_n, S_n)\}$, is a closed set, and $\{\Gamma \in \mathcal{G}_k \mid \Gamma \nrightarrow (G_n, S_n)\}$ is open.

And we have that $w \neq 1$ in $\text{Res}_{\mathcal{C}}((H, S))$ if and only if

$$(H, S) \in \bigcup_{\{n \mid w \neq 1 \text{ in } G_n\}} \{\Gamma \in \mathcal{G}_k \mid \Gamma \rightarrow G_n\}$$

the right hand side condition being a F_{σ} .

And $r = 1$ in $\text{Res}_{\mathcal{C}}((H, S))$ if and only if

$$(H, S) \in \bigcap_{\{n \mid r \neq 1 \text{ in } G_n\}} \{\Gamma \in \mathcal{G}_k \mid \Gamma \nrightarrow G_n\}$$

the right hand side condition being a G_{δ} .

We thus find that the preimage by $\text{Res}_{\mathcal{C}}$ of a basic open set is a finite intersection of F_{σ} and G_{δ} sets, it is thus in Σ_3^0 .

And thus the map $\text{Res}_{\mathcal{C}}$ is indeed Σ_3^0 -measurable. $\square$

Definition 7.5. We say that that $\mathcal{RC}$ is *non-trivial* if it is neither the set $\mathcal{G}$ of all marked groups nor reduced to the trivial group.

Lemma 7.6 (Small cancellation lemma). *Let $\mathcal{C}$ be a set of finitely generated groups. Suppose that $\mathcal{RC}$ is non-trivial. Then:*

$$\text{Res}_{\mathcal{C}} : \mathcal{G}_{\text{WP}} \rightarrow \mathcal{G}_{\text{WP}} \equiv_W \text{Res}_{\mathcal{C}} : \mathcal{G}_{\text{Pres}} \rightarrow \mathcal{G}_{\text{WP}}.$$

Proof. Fix a marked group (H, T) which is not residually $\mathcal{C}$, and an element $w \in H \setminus \{1\}$ which is killed by all morphisms going from H to groups of $\mathcal{C}$.

Let (G, S) be a marked group given by a presentation $\langle S \mid r_1, r_2, \dots \rangle$.

Up to replacing G by $G * \mathbb{Z}$, we can suppose that S contains at least two elements a and b . Let $\mathbb{F}_S$ be the free group on S .

We define a group Γ as a quotient of the free product $\mathbb{F}_S * H$, quotiented by relations $w_i = 1$, $i \geq 5$, where w_i is given by

$$w_i = r_{i-4} w a^{i^2} w a^{-i^2} b^{i^2} w b^{-i^2} a^{i^2+1} w a^{-i^2-1} \dots a^{i^2+2i} w a^{-i^2-2i} b^{i^2+2i} w b^{-i^2-2i} w.$$

We equip Γ with the generating set (S, T) .

We now show:

- (1) The word-problem of $(\Gamma, (S, T))$ can be continuously constructed from the presentation of (G, S) ,
- (2) The word-problem of $(\text{Res}_{\mathcal{C}}(G), S)$ can be continuously constructed from the word-problem of $(\text{Res}_{\mathcal{C}}(\Gamma), (S, T))$.

Those two facts together immediately imply that we have a Weihrauch reduction $\text{Res}_{\mathcal{C}} : \mathcal{G}_{\text{WP}} \rightarrow \mathcal{G}_{\text{WP}} \geq_W \text{Res}_{\mathcal{C}} : \mathcal{G}_{\text{Pres}} \rightarrow \mathcal{G}_{\text{WP}}$, and, the other reduction being trivial, the lemma follows.

We first prove (1).

This relies on the small cancellation condition $C(1/6)$ in free products. We briefly recall some basic facts from [LS77, Section 9, Chapter V].

For an element g of a free product $A * B$, its *length* $|g|$ is the minimal n such that g can be written as a product $x_1 \dots x_n$, where consecutive elements x_i, x_{i+1} do not belong to the same factor.

A product $g = uv$ is in *reduced form* if $|g| = |u| + |v|$, and in *semi-reduced form* if $|g| \geq |u| + |v| - 1$ (i.e. no cancellation occurs in the product, but u could end by an element of the same factor as the starting element of v).

An element b is a *piece* in a set $R \subseteq A * B$ if there are distinct elements r_1 and r_2 in R with $r_1 = bc_1$ and $r_2 = bc_2$, where both products are in semi-reduced form.

A set R satisfies *condition $C'(1/6)$ for free products* if for any r in R , if r can be expressed as a semi-reduced product $r = bc$, where b is a piece for R , then $|b| < |r|/6$.

A set $R \subseteq A * B$ is called *symmetrized* if it is stable by conjugation-reduction.

We will use [LS77, Theorem 9.3]: let R be a symmetrized subset of a free product $A * B$ satisfying the $C'(1/6)$ condition for free products. Then any non-identity element x of the normal closure of R in $A * B$ has a unique expression

$$x = usv,$$

where the product is in reduced form, and there is some r in R such that $r = st$ in reduced form and $|s| > |r|/2$. In words: every relation contains at least half of a relation of one of the elements in R .

Below, we show that the given presentation of Γ satisfies this condition $C'(1/6)$. It follows that the relations of Γ of length less than a given i do not depend on the relations r_j of G for $j \geq i$, which ensures continuity of the map mapping the presentation of G to the word-problem of Γ , and thus (1) holds.

Thus all that is left to establish (1) is to prove that the set of relations $\{w_i\}$ together with their cyclically reduced conjugates satisfy the condition $C'(1/6)$. It is clear that the words w_i are themselves cyclically reduced.

Each word w_i is a product of elements of $\mathbb{F}_S$ together with the element w which belongs to H . Notice that if a word wuw appears both as a subword of w_i and of w_j , for $j \neq i$, it must be that u is one of the relations of the presentation $\langle S \mid r_1, r_2, \dots \rangle$ of G (since the other elements of $\mathbb{F}_S$ that appear in the relations w_i were chosen distinct). And thus a piece has length at most 7: it must have the form

$$xwywzwt,$$

where $(x, y, z, t) \in \mathbb{F}_S$, and y and z are both relations r_i .

Notice that the relation w_i has length $8i + 6$, and thus the shortest relation of Γ , w_5 , has length 46, and $7/46 < 1/6$.

And thus the small cancellation condition holds.

We now prove (2).

In the group $\text{Res}_\mathcal{C}(\Gamma)$, the element w is trivial. Notice that, thanks to $w = 1$, each relation w_i simplifies down to $r_i = 1$. And thus, in the group $\text{Res}_\mathcal{C}(\Gamma)$, the elements of S satisfy the relations of G . It follows that they generate exactly the group $\text{Res}_\mathcal{C}(G)$.

And thus the word problem of $\text{Res}_\mathcal{C}(G)$ can be continuously constructed from the word problem of $\text{Res}_\mathcal{C}(\Gamma)$. $\square$

The above lemma has several useful corollaries.

Corollary 7.7. *Let $\mathcal{C}$ be a set of finitely generated groups. Suppose that $\mathcal{RC}$ is non-trivial. Then:*

$$\text{Res}_\mathcal{C} : \mathcal{G}_{\text{WP}} \rightarrow \mathcal{G}_{\text{WP}} \geq_W \text{Id} : \mathcal{G}_{\text{Pres}} \cap \mathcal{RC} \rightarrow \mathcal{G}_{\text{WP}}.$$

Proof. We have

$$\begin{aligned} \text{Res}_\mathcal{C} : \mathcal{G}_{\text{WP}} \rightarrow \mathcal{G}_{\text{WP}} &\equiv_W \text{Res}_\mathcal{C} : \mathcal{G}_{\text{Pres}} \rightarrow \mathcal{G}_{\text{WP}} \\ &\geq_W \text{Res}_\mathcal{C} : \mathcal{G}_{\text{Pres}} \cap \mathcal{RC} \rightarrow \mathcal{G}_{\text{WP}} \\ &\equiv_W \text{Id} : \mathcal{G}_{\text{Pres}} \cap \mathcal{RC} \rightarrow \mathcal{G}_{\text{WP}}, \end{aligned}$$

where the first equivalence is given by Lemma 7.6, and the reduction $\geq_W$ follows from the fact that the restriction of a problem is always below this problem. $\square$

As a corollary to the above, we get that $\text{Res}_\mathcal{C}$ is always discontinuous.

Corollary 7.8. *Let $\mathcal{C}$ be a set of finitely generated groups. Suppose that $\mathcal{RC}$ is non-trivial. Then $\text{Res}_\mathcal{C}$ is discontinuous, and thus $\text{Res}_\mathcal{C} \geq_W \text{LPO}^* \equiv_W \text{Lim}_{\setminus \omega}$.*

Proof. By Corollary 7.7, it suffices to show that $\text{Id} : \mathcal{G}_{\text{Pres}} \cap \mathcal{RC} \rightarrow \mathcal{G}_{\text{WP}}$ is discontinuous, which amounts to showing that the topology of the space of marked groups is strictly finer than the Scott topology on $\mathcal{RC}$. But $\mathcal{G}_{\text{WP}}$ is metrizable and thus T_2 , whereas the Scott topology is never T_2 in any set that contains a marked group (G, S) and a strict quotient of it, since the constant sequence corresponding to the strict quotient converges to the group (G, S) . $\square$

Another corollary to Lemma 7.6 is the following.

Corollary 7.9. *Let $\mathcal{C}$ be a set of finitely generated groups. If $\mathcal{RC}$ is a quasivariety, then*

$$\text{Res}_\mathcal{C} \equiv_W \text{Id} : \mathcal{G}_{\text{Pres}} \cap \mathcal{RC} \rightarrow \mathcal{G}_{\text{WP}}.$$

Proof. By Corollary 7.7, it suffices to show that $\text{Res}_C \leq_W \text{Id} : \mathcal{G}_{\text{Pres}} \cap \mathcal{RC} \rightarrow \mathcal{G}_{\text{WP}}$. But by Theorem 7.1, because $\mathcal{RC}$ is a quasivariety, the map

$$\text{Res}_C : \mathcal{G}_{\text{WP}} \rightarrow \mathcal{G}_{\text{Pres}}$$

is continuous. The result then follows from the fact that

$$\text{Res}_C : \mathcal{G}_{\text{WP}} \rightarrow \mathcal{G}_{\text{WP}} = (\text{Res}_C : \mathcal{G}_{\text{WP}} \rightarrow \mathcal{G}_{\text{Pres}}) \circ (\text{Id} : \mathcal{G}_{\text{Pres}} \cap \mathcal{RC} \rightarrow \mathcal{G}_{\text{WP}}). \quad \square$$

Proposition 7.10. *Suppose that $\mathcal{RC}$ is not a quasivariety. Then*

$$\text{Res}_C \geq_W \text{LPO}',$$

and thus Res_C is not Σ_2^0 -measurable.

Proof. If $\mathcal{RC}$ is not a quasivariety, then there is a finite set S , an infinite set of relations $R = \{r_0, r_1, r_2, \dots\}$ of $\mathbb{F}_S$, and a relation $w \in \mathbb{F}_S$, such that every group in $\mathcal{RC}$ which satisfies all the relations of R also satisfies w , and yet for each finite subset R_1 of R , there is a group in $\mathcal{RC}$ satisfying the relations of R_1 and in which $w \neq 1$.

We build a reduction to prove that $\text{Res}_C \geq_W \text{LPO}'$. Recall from Example 3.14 that the problem LPO' can be seen as the problem $\exists^\infty$, which maps a sequence of zeroes and ones to 1 if and only if it contains infinitely many ones.

On input a sequence $(u_n)_{n \in \mathbb{N}}$, we build a group presentation $\langle S \mid t_0, t_1, \dots \rangle$, given by $t_i = r_j$, where $j = \#\{n \leq i \mid u_n = 1\}$.

The map $(u_n)_{n \in \mathbb{N}} \mapsto \langle S \mid t_0, t_1, \dots \rangle$ is clearly a continuous map from $\{0, 1\}^{\mathbb{N}}$ to $\mathcal{G}_{\text{Pres}}$.

And by construction $w = 1$ in $\text{Res}_C(\langle S \mid t_0, t_1, \dots \rangle)$ if and only if $\exists^\infty n, u_n = 1$. $\square$

Theorem 7.11. *Let $\mathcal{C}$ be a set of finitely generated groups. Then $\mathcal{RC}$ is a quasivariety if and only if $\text{Res}_C \leq_W \text{Lim}$.*

Proof. By Corollary 7.9, if $\mathcal{RC}$ is a quasivariety, then $\text{Res}_C \equiv_W \text{Id} : \mathcal{G}_{\text{Pres}} \cap \mathcal{RC} \rightarrow \mathcal{G}_{\text{WP}}$. By Lemma 6.3, $\text{Id} : \mathcal{G}_{\text{Pres}} \rightarrow \mathcal{G}_{\text{WP}} \equiv_W \text{Lim}$. Thus $\text{Res}_C \leq_W \text{Lim}$ when $\mathcal{RC}$ is a quasivariety.

And by Proposition 7.10, if $\mathcal{RC}$ is not a quasivariety, then $\text{Res}_C \geq_W \text{LPO}'$, and thus $\text{Res}_C \not\leq_W \text{Lim}$. $\square$

Theorem 7.12. *Let $\mathcal{C}$ be a set of finitely generated groups. The class $\mathcal{C}$ is equationally noetherian if and only if $\text{Res}_C <_W \text{Lim}_{\mathbb{N}}$.*

Proof. Suppose that $\mathcal{C}$ is not equationally noetherian. By Corollary 7.7, $\text{Res}_C \geq_W \text{Id} : \mathcal{G}_{\text{Pres}} \cap \mathcal{RC} \rightarrow \mathcal{G}_{\text{WP}}$, and by Proposition 6.14 $\text{Id} : \mathcal{G}_{\text{Pres}} \cap \mathcal{RC} \rightarrow \mathcal{G}_{\text{WP}} \geq_W \text{Lim}_{\mathbb{N}}$.

Suppose now that $\mathcal{C}$ is equationally noetherian. By Lemma 2.6 and Corollary 7.9 we have $\text{Res}_C \equiv_W \text{Id} : \mathcal{G}_{\text{Pres}} \cap \mathcal{C} \rightarrow \mathcal{G}_{\text{WP}}$, and by Proposition 6.10 we have, for some countable ordinal α , $\text{Id} : \mathcal{G}_{\text{Pres}} \cap \mathcal{C} \rightarrow \mathcal{G}_{\text{WP}} \leq_W \text{Lim}_{\searrow \alpha} <_W \text{Lim}_{\mathbb{N}}$. $\square$

The following immediate corollary provides a gap in terms of Weihrauch complexity for problems of the form Res_C .

Corollary 7.13. *Let $\mathcal{C}$ be a set of finitely generated groups. If $\text{Res}_C \leq_W \text{Lim}_{\mathbb{N}}$, then there exists a countable ordinal α such that $\text{Res}_C <_W \text{Lim}_{\searrow \alpha}$.*

An example of a problem which is below $\text{Lim}_{\mathbb{N}}$ but below no $\text{Lim}_{\searrow \alpha}$ is the problem $\text{Lim}_{\{0,1\}}$, mapping a converging sequence in $\{0, 1\}$ to its limit.

8. BELOW $\text{Lim}_{\mathbb{N}}$: EQUATIONALLY NOETHERIAN SETS

Lemma 2.6, Corollary 7.9 and Theorem 6.10 together yield:

Theorem 8.1. *Let $\mathcal{C}$ be a set of finitely generated groups. Suppose that $\mathcal{C}$ is equationally noetherian. Denote by α the ordinal rank of the lattice $\mathcal{RC}$. If $\alpha \leq \omega^\omega$, then we have the equality:*

$$\text{Res}_{\mathcal{C}} \equiv_W \text{Lim}_{\searrow \alpha}.$$

Otherwise, we have the bounds:

$$\text{Lim}_{\searrow \omega^\omega} \leq_W \text{Res}_{\mathcal{C}} \leq_W \text{Lim}_{\searrow \alpha}.$$

Thus when $\mathcal{C}$ is equationally noetherian, computing the ordinal rank of the lattice $\mathcal{RC}$ is crucial in order to place the problem $\text{Res}_{\mathcal{C}}$ in the Weihrauch lattice.

This rank is always at least ω , as can be seen by using a sequence of morphisms of direct products $G \leftarrow G \times G \leftarrow G \times G \times G \leftarrow \dots$, and using markings with increasingly many generators. Equality is easily seen to be attained for finite sets of finite groups.

Proposition 8.2. *Let $\mathcal{C}$ be a finite set of finite groups. Then $\text{Res}_{\mathcal{C}} \equiv_W \text{Lim}_{\searrow \omega}$.*

Proof. There are only finitely many k -marked residually- $\mathcal{C}$ groups when $\mathcal{C}$ is a finite set of finite groups. Thus the rank of the lattice of residually $\mathcal{C}$ groups is ω , we conclude by Theorem 8.1. $\square$

Let $\mathcal{N}_k$ be the set of finitely generated k -nilpotent groups and $\mathcal{M}$ the set of finitely generated metabelian groups.

Theorem 8.3. *The following hold:*

$$\forall k > 0, \text{Res}_{\mathcal{N}_k} \equiv_W \text{Lim}_{\searrow \omega^2},$$

$$\text{Res}_{\mathcal{M}} \geq_W \text{Lim}_{\searrow \omega^\omega}.$$

In [Cor11], Cornulier asked if ω^ω is the Cantor-Bendixson rank of the space of marked groups. We ask the same question, but for the Scott topology instead:

Problem 8.4. Let $\mathcal{C}$ be an equationally noetherian set of finitely generated groups. Is ω^ω an upper bound to the ordinal rank of the lattice of marked groups in $\mathcal{RC}$?

Another problem is the following:

Problem 8.5. Characterize the Weihrauch complexities of problem of the form $\text{Res}_{\{H\}}$, for H any Gromov hyperbolic group.

8.1. Residually free image. In [Lou12], Louder proves that the rank of the lattice of marked limit groups is ω . Can this result be extended to measure the rank of the lattice of residually free groups?

Problem 8.6. What is the ordinal rank of the lattice of residually free groups?

8.2. Abelianization and p -nilpotent images. The classification of $\text{Res}_{\mathcal{N}_p}$ in the Weihrauch degrees follows from the following proposition.

Proposition 8.7. *Let G be a finitely generated nilpotent group of Hirsch length l . There is $n \in \mathbb{N}$ such that the ordinal rank of the lattice of marked quotients of G is $\omega l + n$.*

Proof. By induction on l , the case $l = 0$ being trivial.

Suppose the result true up to length l .

A nilpotent group of Hirsch length $l + 1$ has finitely many quotients that also have Hirsch length $l + 1$. And thus the rank of the lattice of marked quotient of G can be written as $\alpha + n$, where $n \in \mathbb{N}$ and α is the rank of the lattice of marked quotients of a group H of Hirsch length $l + 1$ which admits no proper quotient of Hirsch length $l + 1$.

By definition, the rank of the lattice of marked quotient of H is the supremum of the ranks of the lattices of its proper quotients plus one, by induction hypothesis each of these has the form $\omega l + p$, and thus their supremum is at most $\omega(l + 1)$. This supremum is in fact attained, as can easily be seen as follows. The center of H must contain a copy of $\mathbb{Z}$ (because an infinite finitely generated nilpotent group has an infinite center). Denote by a a generator of this copy of $\mathbb{Z}$. We get a sequence of marked quotients

$$H/\langle a^2 \rangle \leftarrow H/\langle a^4 \rangle \leftarrow H/\langle a^8 \rangle \leftarrow \dots$$

which shows that the supremum of the ranks of strict quotients of H is indeed $\omega(l + 1)$. $\square$

Corollary 8.8. *For any $p \in \mathbb{N}$, the ordinal rank of the lattice of marked p -nilpotent groups is ω^2 .*

8.3. Metabelian image. The computation of the ordinal length of quotients of metabelian groups relies on the following result of Cornulier:

Theorem 8.9 ([Cor11]). *The Cantor-Bendixson rank of the space of marked metabelian groups is ω^ω .*

By “space of metabelian groups” we mean the subset of the space of marked groups consisting of metabelian groups, equipped with the subset topology.

Because the topology of the space of marked groups is finer than the Scott topology, and because the Scott topology coincides with the Alexandrov topology on the lattice of marked metabelian groups (Proposition 6.9), we get:

Corollary 8.10. *The ordinal rank of the lattice of marked metabelian groups is at least ω^ω .*

9. BETWEEN $\text{Lim}_{\mathbb{N}}$ AND Lim

Theorem 9.1. *Let $\mathcal{C}$ be a set of finitely generated groups. Suppose that $\mathcal{RC}$ is a quasivariety and that it contains a group INIP relative to $\mathcal{RC}$. Then*

$$\text{Res}_{\mathcal{C}} : \mathcal{G}_{\text{WP}} \rightarrow \mathcal{G}_{\text{WP}} \equiv_W \text{Lim}.$$

Proof. By Corollary 7.9 it suffices to show that

$$\text{Id} : \mathcal{G}_{\text{Pres}} \cap \mathcal{RC} \rightarrow \mathcal{G}_{\text{WP}} \equiv_W \text{Lim}.$$

By Lemma 6.4 it suffices to build an injective map $\Psi : \{0, 1\}^{\mathbb{N}} \rightarrow \mathcal{G} \cap \mathcal{RC}$ such that

$$\Psi : \mathbb{S}^{\mathbb{N}} \rightarrow \mathcal{G}_{\text{Pres}}$$

is continuous, and

$$\Psi^{-1} : \mathcal{G}_{\text{WP}} \cap \text{Im}(\Psi) \rightarrow \{0, 1\}^{\mathbb{N}}$$

is also continuous. Let (G, S) be a INIP group relative to $\mathcal{RC}$. Let $\{r_i, i \in \mathbb{N}\} \in \mathbb{F}_k^{\mathbb{N}}$ be a set of relations that are independent relative to $\mathcal{RC}$: for every i_0 there is a

quotient H of $\mathbb{F}_k$ in $\mathcal{RC}$ such that the normal subgroup of H generated by $\{r_i, i \neq i_0\}$ does not contain r_{i_0} .

We define $\Psi : \{0, 1\}^{\mathbb{N}} \rightarrow \mathcal{G} \cap \mathcal{RC}$ by

$$\Psi((u_i)_{i \in \mathbb{N}}) = \text{Res}_{\mathcal{C}}(\mathbb{F}_k / \langle \langle r_i, u_i = 1 \rangle \rangle).$$

The fact that the set $\{r_i, i \in \mathbb{N}\}$ is independent relative to $\mathcal{RC}$ guarantees that Ψ is injective.

The fact that $\Psi : \mathbb{S}^{\mathbb{N}} \rightarrow \mathcal{G}_{\text{Pres}}$ is continuous follows from the fact that $\mathcal{RC}$ is a quasivariety: Ψ is the composition of $\text{Res}_{\mathcal{C}} : \mathcal{G}_{\text{Pres}} \rightarrow \mathcal{G}_{\text{Pres}}$, which is continuous because $\mathcal{RC}$ is a quasivariety (Proposition 7.1), with $(u_i)_{i \in \mathbb{N}} \mapsto \mathbb{F}_k / \langle \langle r_i, i \in A \rangle \rangle$, which is obviously a continuous function between $\mathbb{S}^{\mathbb{N}}$ and $\mathcal{G}_{\text{Pres}}$.

Is is easy to see that $\Psi^{-1} : \mathcal{G}_{\text{WP}} \cap \text{Im}(\Psi) \rightarrow \{0, 1\}^{\mathbb{N}}$ is continuous. Indeed, the ρ_{WP} -name of a marked group (G, S) is the sequences $(u_i)_{i \in \mathbb{N}} \in \{0, 1\}^{\mathbb{N}}$ given by $u_i = 1$ iff $\theta_k(i) = 1$ in (G, S) , where $\#S = k$. Then, $\Psi^{-1}((G, S))$ is the sub-sequence $(v_i)_{i \in \mathbb{N}}$ of $(u_i)_{i \in \mathbb{N}}$ given by

$$v_i = u_{\theta_k^{-1}(r_i)}.$$

The extraction map $(u_i)_{i \in \mathbb{N}} \mapsto (v_i)_{i \in \mathbb{N}}$ is continuous. $\square$

Corollary 9.2. *Let $\mathcal{C}$ be any of:*

- *the set of finitely generated k -solvable groups, for $k > 2$,*
- *the set of finitely generated exponent k groups, for k a big enough odd number,*
- *the set of finitely generated LEF groups,*
- *the set of finitely generated torsion free groups,*
- *the set of finitely generated left orderable groups.*

Then

$$\text{Res}_{\mathcal{C}} : \mathcal{G}_{\text{WP}} \rightarrow \mathcal{G}_{\text{WP}} \equiv_W \text{Lim}.$$

Proof. For each $\mathcal{C}$, it suffices to exhibit a group which is INIP relative to the quasivariety.

For finitely generated k -solvable groups, finitely generated torsion free groups, we can use P. Hall's central-by-metabelian group whose center is $\bigoplus_{i \in \mathbb{N}} \mathbb{Z}$ [Hal54]. See Section 10.2.

This group also works for left orderability. Indeed, an extension of a left-orderable group by a left-orderable group is left orderable [Cla16, Problem 1.8]. And thus any torsion free central quotient of Hall's group remains left-orderable.

For LEF groups, we can use Dyson's doubles of lamplighter groups from [Dys74], see Proposition 10.10.

For groups of finite, big enough, odd exponent, see [AA18], where groups of finite exponent whose center is infinite dimensional are constructed. $\square$

10. FUNCTIONS EQUIVALENT TO Lim'

Here we establish:

Theorem 10.1. *Let $\mathcal{C}$ be any of:*

- *the set of finite groups,*
- *the set of finitely generated nilpotent groups,*
- *the set of finitely presentable groups.*

Then

$$\text{Res}_C : \mathcal{G}_{\text{WP}} \rightarrow \mathcal{G}_{\text{WP}} \equiv_W \text{Lim}'.$$

The proofs are via direct reductions and do not rely on previous lemmas.

We use the construction used to prove Theorem 10.1 to also answer a problem from [BK22]:

Proposition 10.2. *The set $\mathcal{R}\text{Fin}$ of residually finite groups is Π_3^0 complete in the Borel hierarchy on the space of marked groups. So are the set of residually nilpotent and residually finitely presented groups.*

10.1. Residually finite image. We use a construction of Dyson from [Dys74] to classify Res_{Fin} , where Fin is the set of finite groups.

10.1.1. Dyson's doubles of the lamplighter group. The lamplighter group L is the wreath product of $\mathbb{Z}/2$ and $\mathbb{Z}$, noted $\mathbb{Z}/2 \wr \mathbb{Z}$, which is by definition the semi-direct product $\bigoplus_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z} \rtimes \mathbb{Z}$, where $\mathbb{Z}$ acts on $\bigoplus_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z}$ by permuting the indices. It admits the following presentation:

$$\langle a, t \mid a^2, [a, t^i a t^{-i}], i \in \mathbb{Z} \rangle$$

The element $u_i \stackrel{\text{def}}{=} t^i a t^{-i}$ of L corresponds to the element of $\bigoplus_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z}$ with only one non-zero coordinate in position $i \in \mathbb{Z}$. Consider another copy $\hat{L}$ of the lamplighter group, together with an isomorphism from L to $\hat{L}$ we note $g \mapsto \hat{g}$. For each subset $\mathcal{A}$ of $\mathbb{Z}$, define $L(\mathcal{A})$ to be the amalgamated product of L and $\hat{L}$, with $u_i = t^i a t^{-i}$ identified with $\hat{u}_i = \hat{t}^i \hat{a} \hat{t}^{-i}$ for each i in $\mathcal{A}$. It has the following presentation:

$$\langle a, \hat{a}, t, \hat{t} \mid a^2, \hat{a}^2, [a, t^i a t^{-i}], [\hat{a}, \hat{t}^i \hat{a} \hat{t}^{-i}], i \in \mathbb{Z}, t^j a t^{-j} = \hat{t}^j \hat{a} \hat{t}^{-j}, j \in \mathcal{A} \rangle.$$

Recall that the profinite topology on $\mathbb{Z}$ is defined as follows: U is open if for any $z \in U$ there is $k \in \mathbb{Z}$ such that $z + k\mathbb{Z} \subseteq U$. For $\mathcal{A} \subseteq \mathbb{Z}$, we denote by $\overline{\mathcal{A}}$ its closure in the profinite topology.

Proposition 10.3 (Dyson, [Dys74], Theorem 1). *For any $\mathcal{A} \subseteq \mathbb{Z}$, we have*

$$\text{Res}_{\text{Fin}}(L(\mathcal{A})) = L(\overline{\mathcal{A}}).$$

In particular, $L(\mathcal{A})$ is residually finite if and only if $\mathcal{A}$ is closed in the profinite topology on $\mathbb{Z}$.

The following result is immediate:

Proposition 10.4. *The map*

$$\begin{aligned} L : \mathcal{P}(\mathbb{Z}) &\rightarrow \mathcal{G} \\ A &\mapsto L(A) \end{aligned}$$

is injective, and it defines a homeomorphism onto its image when $\mathcal{P}(\mathbb{Z})$ and $\mathcal{G}$ have either their Scott topologies ($\mathbb{S}^{\mathbb{Z}}$ and $\mathcal{G}_{\text{Pres}}$), or their Polish topologies ($\{0,1\}^{\mathbb{Z}}$ and $\mathcal{G}_{\text{WP}}$). $\square$

As a corollary of the above, we have a reduction: Res_{Fin} reduces to the closure map $C : \mathcal{P}(\mathbb{Z}) \rightarrow \mathcal{P}(\mathbb{Z}), A \mapsto \overline{A}$.

Corollary 10.5. *There is a Weihrauch reduction*

$$C \leq_W \text{Res}_{\text{Fin}}.$$

Proof. By Proposition 10.4, we can introduce L^{-1} , which is a continuous map from the space of marked groups to $\mathcal{P}(\mathbb{Z})$ with the Cantor space topology. The reduction is then given by: $C(\mathcal{A}) = L^{-1}(\text{Res}_{\text{Fin}}(L(\mathcal{A})))$ for any $\mathcal{A} \subseteq \mathbb{Z}$. $\square$

10.1.2. Classification of Res_{Fin} .

Lemma 10.6. *We have the Weihrauch equivalence: $C \equiv_W \text{Lim}'$.*

Proof. We first remark that $\mathbb{Z}$ equipped with the profinite topology is homeomorphic to $\mathbb{Q}$ equipped with the Euclidian topology: by classical results of Sierpinski and Frechet, $\mathbb{Q}$ is, up to homeomorphism, the only nonempty, countable, metrizable and perfect space (see [Kec95, 7.12]). Thus C is equivalent to the closure map on $\mathbb{Q}$, which we denote by $C_{\mathbb{Q}}$:

$$C_{\mathbb{Q}} : \mathcal{P}(\mathbb{Q}) \rightarrow \mathcal{P}(\mathbb{Q}), A \mapsto \overline{A}.$$

(The representation of $\mathbb{Q}$ that we use is associated to the discrete topology, where $\pm \frac{p}{q}$ is given by the triple $(\pm 1, p, q)$.)

The problem Lim' is equivalent to the infinite parallelization of the problem LPO' [BGP21]. In other words, the input is a double sequence $(u_{n,i})_{(n,i) \in \mathbb{N}^2}$ of digits in $\{0, 1\}$, and its output is the sequence $(v_i)_{i \in \mathbb{N}} \in \{0, 1\}^{\mathbb{N}}$ given by

$$v_i = 1 \iff \exists^\infty n, u_{n,i} = 1.$$

We first show that $C_{\mathbb{Q}} \geq_W \text{Lim}'$.

For a double sequence $(u_{n,i})_{(n,i) \in \mathbb{N}^2}$, define a subset A of $\mathbb{Q}$ by

$$i + \frac{1}{n+2} \in A \iff u_{n,i} = 1.$$

It follows immediately that $i \in \overline{A} \iff \exists^\infty n, u_{n,i} = 1$. This provides the desired reduction.

Conversely, we show that $\text{Lim}' \geq_W C_{\mathbb{Q}}$.

Consider $A \subseteq \mathbb{Q}$ and $r \in \mathbb{Q}$. We then have

$$r \in \overline{A} \iff (\forall n \in \mathbb{N} \exists q \in A, |q - r| < 2^{-n}).$$

This $\forall \exists$ statement can be converted to an instance of LPO' : we define a double sequence $d_{n,p}$ given by $d_{n,p} = 1$ if there exists a rational q with numerator and denominator both at most p which is in A and at distance at most 2^{-n} of r , and if furthermore $d_{n,t} = 0$ for $t < p$. We then have

$$r \in \overline{A} \iff \exists^\infty (n, p), d_{n,p} = 1$$

It follows that $C_{\mathbb{Q}}$ is computed by the parallelization of LPO' , which is precisely Lim' . $\square$

Corollary 10.7. *We have the equivalence*

$$\text{Lim}' \equiv_W \text{Res}_{\text{Fin}}.$$

Proof. The reduction $\text{Res}_{\text{Fin}} \leq_W \text{Lim}'$ follows from Proposition 7.4.

The fact that $\text{Lim}' \leq_W \text{Res}_{\text{Fin}}$ follows from $C \equiv_W \text{Lim}'$ (Lemma 10.6) and $C \leq_W \text{Res}_{\text{Fin}}$ (Corollary 10.5). $\square$

10.1.3. *Relation with the residual finiteness growth.* Let (G, S) be a residually finite marked group. The *residual finiteness growth* of (G, S) [BR10] is the function which maps n to the smallest k such that every non-trivial element of length at most n in (G, S) has a non trivial image in a finite quotient of G of size at most k .

Groups with arbitrarily large residual finiteness growth were constructed in [BRS16]. Here we note that the proof of the fact that $\text{Lim}' \equiv_W \text{Res}_{\text{Fin}}$ necessarily relies on arbitrarily rapid residual finiteness growth.

Note that the construction of Neumann used in [BRS16] produces only residually finite groups. Thus this construction cannot be used to prove $\text{Lim}' \equiv_W \text{Res}_{\text{Fin}}$. Another group construction where arbitrary residual finiteness growth is obtained, but which does not produce only residually finite groups, is given in [DR25], this construction could probably be used to establish the equivalence $\text{Lim}' \equiv_W \text{Res}_{\text{Fin}}$.

Proposition 10.8. *Consider the restriction $(\text{Res}_{\text{Fin}})_{|\mathcal{A}}$ of the problem Res_{Fin} to a set $\mathcal{A}$ of groups, such that there is a universal asymptotic upper bound f to the residual finiteness growth of the groups in $\text{Res}_{\text{Fin}}(\mathcal{A})$.*

*Then $(\text{Res}_{\text{Fin}})_{|\mathcal{A}} \leq_W \text{Lim}_{\mathbb{N}} * \text{Lim} = \text{Lim}'_{\mathbb{N}}$, and thus $(\text{Res}_{\text{Fin}})_{|\mathcal{A}} <_W \text{Lim}'$.*

Proof. Let (G, S) be a marked group in $\mathcal{A}$.

For each n we try to build the ball of radius n of the Cayley graph of $\text{Res}_{\text{Fin}}((G, S))$.

We act as though f was an actual upper bound to the residual finiteness growth of (G, S) (and not only an asymptotic upper bound).

List all finite marked groups of size at most $f(n)$. By applying LPO to each of them, it is possible to determine exactly those that are marked quotients of (G, S) . We then build a finite graph Γ_n by saying that w is non-trivial in Γ_n if and only if it has a non trivial image in one of the listed marked finite groups.

By doing the above in parallel for each n , we get a sequence $\Gamma_0, \Gamma_1, \Gamma_2 \dots$ of finite graphs. This sequence can be obtained by solving in parallel infinitely many instances of LPO, i.e., by applying Lim . Because f is an asymptotic upper bound to the residual finiteness growth of (G, S) , there is some N such that each Γ_k , $k \geq N$, is actually the ball of radius k in the Cayley graph of (G, S) .

Note that this N is reached exactly when contradictions stop arising between the graphs Γ_i :

$$N = \inf\{k \mid \forall n \geq k, \Gamma_{n+1} \text{ properly extends } \Gamma_n\}.$$

Define a sequence $(u_n)_{n \in \mathbb{N}}$ by $u_{n+1} = u_n$ if Γ_{n+1} properly extends Γ_n , and $u_{n+1} = u_n + 1$ otherwise. This sequence depends continuously on the sequence $(\Gamma_n)_{n \in \mathbb{N}}$, and it converges. One application of $\text{Lim}_{\mathbb{N}}$ can provide the limit of this sequence. And the sequence $(\Gamma_n)_{n \in \mathbb{N}}$ together with this limit provide exactly the Cayley graph of $\text{Res}_{\text{Fin}}((G, S))$. $\square$

10.1.4. *Two more uses of Dyson's construction.* Here we prove two additional results based on Dyson's construction: there exists a group INIP relative to the quasivariety of LEF groups (a fact used in Corollary 9.2), and that the set of residually finite groups is Π_3^0 -complete in the space of marked groups (a question left open in [BK22], where the upper bound Π_4^0 was given).

Proposition 10.9. *For every $A \subseteq \mathbb{Z}$, the group $L(A)$ is LEF.*

Proof. If A is closed in the profinite topology on $\mathbb{Z}$, then $L(A)$ is even residually finite. If A is not closed, A is the limit, in the Cantor space topology on $\mathcal{P}(\mathbb{Z})$,

of a sequence of closed sets: take for instance finite subsets of A that exhaust it. Because L is continuous, $L(A)$ is the limit in $\mathcal{G}_{\text{WP}}$ of a sequence of residually finite groups, thus it is LEF. $\square$

Proposition 10.10. *The group $L(\emptyset)$ is INIP relative to the set of LEF groups.*

Proof. Consider an infinite subset A of $\mathbb{Z}$ such that for every $B \subseteq A$, B is closed in the profinite topology of $\mathbb{Z}$. For instance A could be the set of primes. In this case, the relations $u_i = \hat{u}_i$, $i \in A$, are independent modulo the set of LEF groups. Indeed, for each $i \in A$, the group $L(A \setminus \{i\})$ is LEF, and $u_i = \hat{u}_i$ is not a relation in it. $\square$

Proposition 10.11. *The set of residually finite groups is Π_3^0 -complete in the space of marked groups.*

Proof. Being residually finite is in Π_3^0 . A marked group (G, S) is residually finite if for any non-trivial element $w \in G \setminus \{1\}$ there is some finite marked group (F, S') in which $w \neq 1$ such that every relation of (G, S) is a relation of (F, S') :

$$\begin{aligned} \forall w \in \mathbb{F}_S, \exists (F, S') \text{ finite}, \forall r \in \mathbb{F}_S, \\ w \neq_{(G, S)} 1 \implies (w \neq_{(F, S')} 1 \ \& \ (r =_{(G, S)} 1 \implies r =_{(F, S')} 1)). \end{aligned}$$

We now prove completeness.

Via Dyson's construction, we get that $L(A)$ is residually finite if and only if A is closed in the profinite topology on $\mathbb{Z}$. Recall that $\mathbb{Z}$ equipped with the profinite topology is homeomorphic to $\mathbb{Q}$ with the subset topology of $\mathbb{R}$ (proof of Lemma 10.6). We thus get a Wadge reduction between the set of residually finite groups and the set of closed subsets of $\mathbb{Q}$ (in the Polish space of subsets of $\mathbb{Q}$ equipped with the prodiscrete topology).

And thus it suffices to show that being closed in the Euclidian topology of $\mathbb{Q}$ is a Π_3^0 complete property.

We Wadge reduce the Π_3^0 -complete set

$$S_3 = \{x \in 2^{\mathbb{N} \times \mathbb{N}} \mid \forall n, \forall^\infty m, x(n, m) = 0\}$$

to the set of closed subsets of $\mathbb{Q}$. For $x \in 2^{\mathbb{N} \times \mathbb{N}}$, we define a set of rationals A_x via

$$n + \frac{1}{m+1} \in A_x \iff x(n, m) = 1.$$

It is easy to see that the map $x \mapsto A_x$ is continuous, and that A_x is closed if and only if $x \in S_3$. $\square$

10.2. Hall's central-by-metabelian group. The following construction due to P. Hall [Hal54] will be useful for both the classification of the maps associated to residually nilpotent and residually finitely presented images.

We first describe it. In what follows, we use the convention that $[x, y] = x^{-1}y^{-1}xy$, and define inductively $[x, y, z] = [[x, y], z]$, $[x, y, z, v] = [[x, y, z], v]$ and so on. Furthermore we will write x^y for $y^{-1}xy$.

Hall's group G is generated by two elements a and b . For i in $\mathbb{Z}$, denote by b_i the element $a^{-i}ba^i$. The group G is then given by the presentation

$$\pi_1 = \langle a, b \mid [b_i, b_j, a] = 1, [b_i, b_j, b] = 1, \forall i, j \in \mathbb{Z} \rangle.$$

Note that $[b_i, b_j]^a = [b_{i+1}, b_{j+1}]$, so

$$[b_i, b_j, a] = 1 \iff [b_i, b_j] = [b_{i+1}, b_{j+1}],$$

and thus the above presentation could be written equivalently as

$$\pi'_1 = \langle a, b \mid [b, b_i, a] = 1, [b, b_i, b] = 1, \forall i \in \mathbb{Z} \rangle.$$

The group G can both be seen as a maximal central extension of the wreath product $\mathbb{Z} \wr \mathbb{Z}$ and as an HNN extension of an infinitely generated two step nilpotent group generated by the elements $b_i, i \in \mathbb{Z}$. The group G is thus center-by-metabelian and (class 2 nilpotent)-by-abelian.

Denote by d_i the element $[b_0, b_i]$, for $i \in \mathbb{Z}$. Then $d_i = d_{-i}^{-1}$ and $d_i = [b_t, b_{t+i}]$ for all $t \in \mathbb{Z}$.

The center of G is a free abelian group on countably many generators [Hal54], in fact the elements $d_i, i > 0$, form the basis of the center of G .

In [BPR25], it was shown that, to consider nilpotent quotients of G , using another basis for the center is useful. Consider the element f_n given by

$$f_n = [b, \underbrace{a, \dots, a}_n, b].$$

Denote by $\gamma_n(G)$ the n -th term in the lower central series of G , given by $\gamma_1(G) = G$ and $\gamma_{n+1}(G) = [\gamma_n(G), G]$. A group H is residually nilpotent if and only if

$$\bigcap_{n \geq 1} \gamma_n(H) = \{1\}.$$

Lemma 10.12 ([BPR25, Lemma 68, Lemma 70]). *In the notations introduced above:*

- (1) *The elements f_n form another basis of the center of G ;*
- (2) *Each $f_n \in \gamma_{n+2}(G)$;*
- (3) *And G is residually nilpotent, i.e., there is some function² h s.t. $f_n \in \gamma_{n+2}(G) \setminus \gamma_{h(n)}(G)$.*

It follows that a central quotient of G obtained by adding relations of the form $f_i = f_j$ is residually nilpotent if and only if, for each i , f_i is identified to only finitely many $f_j, j \neq i$.

10.3. Residually nilpotent image. By Proposition 7.4, $\text{Res}_{\mathcal{N}} \leq_W \widehat{\text{LPO}}'$.

Lemma 10.13. *Let $\mathcal{N}$ be the set of finitely generated nilpotent groups. Then $\text{Res}_{\mathcal{N}} \geq_W \widehat{\text{LPO}}'$.*

Proof. A double sequence $u = (u_{n,m}) \in \{0,1\}^{\mathbb{N}^2}$ is mapped, via $\widehat{\text{LPO}}'$, to the sequence $(v_n) \in \{0,1\}^{\mathbb{N}}$ given by

$$v_n = 1 \iff \exists^\infty m, u_{n,m} = 1.$$

Associated to the double sequence u , we consider a central quotient of Hall's group G , obtained by adding to it the following relations:

$$f_{\langle n,0 \rangle} = f_{\langle n,m \rangle} \text{ for each } m \text{ s.t. } u_{n,m} = 1.$$

(Recall that $n, m \mapsto \langle n, m \rangle$ is Cantor's pairing function.) Denote by G_u the obtained group. It is easy to see that the word-problem of G_u can be continuously constructed from the sequence u . And:

²In fact, for n odd $f_n \in \gamma_{n+2}(G) \setminus \gamma_{n+3}(G)$, but for n even no precise bound for h was computed in [BPR25].

- If there are only finitely many m with $u_{n,m} = 1$, $f_{\langle n,0 \rangle}$ has a non-trivial image in a nilpotent quotient of G_u .
- If there are infinitely many m with $u_{n,m} = 1$, $f_{\langle n,0 \rangle} \in \gamma_m(G)$ for every m , and thus $f_{\langle n,0 \rangle}$ is trivial in the residually nilpotent image of G .

It follows from this that $f_{\langle n,0 \rangle} = 1$ in $\text{Res}_{\mathcal{N}}((G_u, (a, b)))$ if and only if $\exists^\infty m, u_{n,m} = 1$. This gives the desired reduction. $\square$

Proposition 10.14. *The set of residually nilpotent groups is Π_3^0 -complete in the space of marked groups.*

Proof. The proof is identical to that of Proposition 10.11, replacing the use of Dyson's doubles of the lamplighter group by central quotients of Hall's group, as above. $\square$

10.4. Residually finitely presented image. Because of Higman's Embedding Theorem, we rely on notions coming from computability theory to classify $\text{Res}_{\mathcal{FP}}$, where $\mathcal{FP}$ is the set of finitely presented groups.

Lemma 10.15. *Let $\mathcal{FP}$ be the set of finitely presented groups. Then $\text{Res}_{\mathcal{FP}} \geq \widehat{\text{LPO}}'$.*

Proof. Consider, as input of $\widehat{\text{LPO}}'$, a sequence $u = (u_{n,m}) \in \{0,1\}^{\mathbb{N}^2}$, we must construct a sequence $(v_n) \in \{0,1\}^{\mathbb{N}}$ with

$$v_n = 1 \iff \exists^\infty m, u_{n,m} = 1.$$

We again consider a central quotient of Hall's group G , using the basis $(f_n)_{n \in \mathbb{N}}$ of the center which is appropriate to study nilpotent quotients (see Section 10.2).

Let $A \subseteq \mathbb{N}$ be a simple set, i.e. a c.e. set whose complement, while infinite, does not contain an infinite c.e. set.

We suppose $0 \notin A$. Let $a_0 \leq a_1 \leq a_2 \dots$ be the (non-computable) enumeration of the complement of A .

For every $n \in \mathbb{N}$ and $m \in A$, we add the relation $f_{\langle n,m \rangle} = 1$ to G .

For every $n \in \mathbb{N}$ and $m \in \mathbb{N}$, if $u_{n,m} = 1$, we add the relation $f_{\langle n,0 \rangle} = f_{\langle n,a_m \rangle}$.

Denote by G_u the obtained group. It is easy to see that the word-problem of G_u can be continuously constructed from the sequence u .

And:

- If there are only finitely many m with $u_{n,m} = 1$, $f_{\langle n,0 \rangle}$ has a non-identity image in a nilpotent quotient of G_u (see Section 10.3), this quotient is finitely presentable.
- If there are infinitely many m with $u_{n,m} = 1$, an infinite number of relations of the form $f_{\langle n,0 \rangle} = f_{\langle n,a_m \rangle}$ were added. In any recursively presented quotient of G_u (and in particular any finitely presented quotient), the set $\{p \mid f_{\langle n,0 \rangle} = f_{\langle n,p \rangle}\}$ is a c.e. set. Thus it cannot be contained in the complement of A , and it must meet A , and thus $f_{\langle n,0 \rangle} = 1$ in this group.

It follows from this that $f_{\langle n,0 \rangle} = 1$ in the residually finitely presented image of G_u if and only if $\exists^\infty m, u_{n,m} = 1$. This gives the desired reduction. $\square$

Proposition 10.16. *The set of residually finitely presented groups is Π_3^0 -complete in the space of marked groups.*

Proof. The proof is identical to that of Proposition 10.11, replacing the use of Dyson’s doubles of the lamplighter group by central quotients of Hall’s group, as above. $\square$

11. ABOVE $\text{Lim}'_{\mathbb{N}\mathbb{N}}$

Proposition 7.3 provides examples of sets $\mathcal{C}$ such that $\text{Res}_{\mathcal{C}}$ is not measurable, and thus in particular not Π^0_3 -measurable.

It would be interesting to classify precisely some problems $\text{Res}_{\mathcal{C}}$ that are strictly above $\text{Lim}'_{\mathbb{N}\mathbb{N}}$. We leave the following as a problem.

Problem 11.1. Prove that, if $\mathcal{C}$ is one of: the set of finitely generated solvable groups, the set of finitely generated torsion groups, then $\text{Res}_{\mathcal{C}}$ is not Π^0_3 -measurable.

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